

Kindergarten

Volume II: Real Analysis

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Preface

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Preliminaries

0.1 Basic notations

Let X be a set and $\mathbf{P}_j(x)$, $j = 1, 2, \dots, n$ propositions constraint to $x \in X$, then we sometimes simply write $\{\mathbf{P}_1(x), \dots, \mathbf{P}_n(x)\}$ instead of $\{x : \mathbf{P}_1(x), \dots, \mathbf{P}_n(x)\}$. If a proposition $\mathbf{P}(x)$ is $x \in A$ for some subset $A \in X$, then we write $\{A, \mathbf{P}_1(x), \dots, \mathbf{P}_n(x)\}$ simply instead of $\{x : x \in A, \mathbf{P}_1(x), \dots, \mathbf{P}_n(x)\}$, and so on. For example, if $f : X \rightarrow \mathbb{R}$ is a function then $\{f > a\}$ means $\{x : f(x) > a\}$, and $\{A, B, f \leq b\}$ means $\{x : x \in A, x \in B, f(x) \leq b\}$.

However, if we write $\{A, B\}$ then it can be interpret either a doubleton containing sets A and B , or the set $\{x : x \in A, x \in B\}$. In this case, we always consider the former one because there is no proposition constraint to x .

0.2 Orderings

[Explain why we introduce an ordering in rigor]

We can consider two different kind of orderings. One is a generalization of $\leq$ on real numbers, and the other is the one of $<$ on real numbers.

Definition 0.2.1 (Ordering)

Let X be a set. A relation R in X is an (*partial*) *ordering* if it satisfies the following properties.

- (i) aRa for every $a \in X$.
- (ii) For $a, b \in X$, if aRb and bRa , then $a = b$.
- (iii) For $a, b, c \in X$, if aRb and bRc , then aRc .

Then we say (X, R) or simply X an *ordered set*.

Definition 0.2.2 (Strict Ordering)

Let X be a set. A relation S in X is a *strict ordering* if it satisfies the following conditions.

- (i) If aSb holds, then bSa is not true.
- (ii) For $a, b, c \in X$, if aSb and bSc , then aSc .

For example, one can easily check that $\leq$ is an order in the real number system, and $<$ is a strict order in the real number system. Notice that $a \leq b$ holds if and only if either $a = b$ or $a < b$ holds. On the other hand, $a < b$ if and only if $a \leq b$ and $a \neq b$. Those relations also true for orderings and strict orderings. The following proposition is Theorem 2.5.6 in [3].

Proposition 0.2.3

(a) Let R be an ordering of X . Then the relation S defined in X by

$$aSb \text{ if and only if } aRb \text{ and } a \neq b$$

is a strict order of A .

(b) Let S be a strict order of X . Then the relation R defined in X by

$$aRb \text{ if and only if } aSb \text{ or } a = b$$

is an ordering of X .

proof. See the proof of Theorem 2.5.6 in [3]. □

Remark 0.2.4

Throughout of this book, we will use a symbol $\leq$ as an order on X , and $<$ a strict order on X , in general. In addition, if one defines an order $\leq$, then $<$ is always understood in the sense of Proposition 0.2.3, and vice versa.

Definition 0.2.5 (Total Ordering, Well Ordering)

If an ordering is called by *totally ordered* if

(iv) For $a, b \in X$, either $a < b$, $a = b$, or $a > b$ is satisfied.

In this case, we say that an ordered set as a *totally ordered set*.

Finally, if a totally ordered set has a property that every nonempty set has the minimum element, we say that the totally ordered set is *well ordered*.

Definition 0.2.6

Let $(X, \leq)$ be an ordered set. Then we define

$$\begin{aligned} (a, b) &:= \{x : a < x < b\}, & [a, b] &:= \{x : a \leq x \leq b\}, \\ (a, b] &:= \{x : a < x \leq b\}, & [a, b) &:= \{x : a \leq x < b\}. \end{aligned}$$

All of four sets are called an *interval*. Especially, (a, b) is called an *open interval* and $[a, b]$ is called a *closed interval*.

0.3 Product of sets

Definition 0.3.1 (Product set)

Let A be a nonempty set and for each $\alpha \in A$ consider some *nonempty* set X_α . Then define a *product* $\prod_{\alpha \in A} X_\alpha$ of sets $\{X_\alpha\}_{\alpha \in A}$ by a collection of all functions $f : A \rightarrow \bigcup_{\alpha \in A} X_\alpha$ such that $f(\alpha) \in X_\alpha$ for each $\alpha \in A$.

If at least one set X_α is empty or $A = \emptyset$, define $\prod_{\alpha \in A} X_\alpha := \emptyset$.

In mathematics, if every sets are nonempty, their product is also not empty and such fact is an axiom. Formally,

Axiom 0.3.2 (Axiom of choice)

Let A be a nonempty set and for each $\alpha \in A$ consider some *nonempty* set X_α . Then

$$\prod_{\alpha \in A} X_\alpha \neq \emptyset.$$

0.4 Ordinals

On the set theory (see [3] for instance), one defines natural numbers by following:

$$0 := \emptyset, \quad 1 := \{0\}, \quad 2 := \{0, 1\}, \quad 3 := \{0, 1, 2\}, \dots$$

For their definitions, if n is a natural number, we can observe the following facts, their rigorous proofs is written in [3].

- (a) For any natural number n , every element of n is a subset of n . For example, notice that $2 \in 3$ by the definition of 3. Since $2 = \{0, 1\}$ and $3 = \{0, 1, 2\}$, we have $2 \subset 3$.
- (b) The set n together with a strict order $\in$ is well-ordered. Recall that then we define an order $\leq$ in the sense of Remark 0.2.4.

Thus one can generalize natural numbers by the following way.

Definition 0.4.1 (Ordinal Numbers)

A set α is an *ordinal number* if

- (a) Every element of α is a subset of α .
- (b) $\in$ is a strict order of α which is well-ordered.

We define a *successor ordinal* of α by $\alpha + 1 := \alpha \cup \{\alpha\}$.

Remark 0.4.2

By the definition of natural numbers and ordinal numbers, every natural number is an ordinal number.

User Guides for General Topology

1.1 Basic Terminologies for Topology

In Volume 1, we explored basic terminologies and their properties of metric spaces. In this chapter, we will extend those by introducing the concept called “topology”.

Definition 1.1.1

Let X be a set. A *topology* is a collection τ of some subsets of X satisfies the following conditions.

- (i) $\emptyset, X \in \tau$;
- (ii) If $\mathcal{U} \subset \tau$, then $\bigcup \mathcal{U} \in \tau$;
- (iii) If $\mathcal{F} \subset \tau$ be a finite subcollection, then $\bigcap \mathcal{F} \in \tau$.

Every element of a topology is called an *open set*, and complement of open sets are called *closed sets*.

Everywhere in this book, **topological spaces are not empty**. Below exercise gives a tool whether a set is an open set. We will use the following exercise without mentioning afterwords.

Exercise 1.1.2*

Let X be a topological space and E be a nonempty subset of X such that for every $x \in E$, there exists an open set U such that $x \in U \subset E$. Prove that E is open in X .

By the definition of closed sets, we can clearly obtain the following.

Proposition 1.1.3

- (i) If $\mathcal{U}$ is a collection of some closed sets, then $\bigcap \mathcal{U}$ is also closed.
- (ii) If $\mathcal{F}$ is a finite collection of some closed sets, then $\bigcup \mathcal{F}$ is closed.

If X is a nonempty set, one can define several topologies on X . Let us consider a collection $\mathcal{T}$ of some topologies of X . Then interestingly, the intersection $\bigcap \mathcal{T}$ is also a topology of X . Here is the proposition for it.

Proposition 1.1.4

Let X be a nonempty set and $\mathcal{T}$ a collection of some topologies of X . Then $\bigcap \mathcal{T}$ is also a topology of X .

proof. Since each element of $\mathcal{T}$ is a topology of X , clearly $\emptyset$ and X belongs in $\bigcap \mathcal{T}$. If $\mathcal{U} \subset \bigcap \mathcal{T}$, then $\mathcal{U} \subset \tau$ for every $\tau \in \mathcal{T}$. Since τ is a topology, $\bigcup \mathcal{U} \in \tau$ for all $\tau \in \mathcal{T}$, hence $\bigcup \mathcal{U} \in \bigcap \mathcal{T}$. Similarly, if $\mathcal{F}$ is a finite subcollection of $\bigcap \mathcal{T}$, one can prove that $\bigcap \mathcal{F} \in \bigcap \mathcal{T}$. Therefore, $\bigcap \mathcal{T}$ is a topology. □

Consider a collection $\mathcal{E}$ of some subsets of X and $\mathcal{T}$ a collection of all topologies which contains $\mathcal{E}$. Since $\mathcal{P}(X)$ is clearly a topology and $\mathcal{E} \subset \mathcal{P}(X)$, the collection $\mathcal{T}$ is not empty. Thanks to Proposition 1.1.4, $\bigcap \mathcal{T}$ is a topology of X , and $\bigcap \mathcal{T} \supset \mathcal{E}$. By the definition of $\mathcal{T}$, $\bigcap \mathcal{T}$ is contained in every topology which contains $\mathcal{E}$. For this perspective, we can say that $\bigcap \mathcal{T}$ is the smallest topology which contains $\mathcal{E}$. Thus the following definition makes sense.

Definition 1.1.5 (Subbasis)

Let X be a nonempty set and $\mathcal{E}$ a collection of some subsets of X . Then the smallest topology which contains $\mathcal{E}$ exists. We say that topology as a one *generated by* $\mathcal{E}$, and is denoted by $\tau(\mathcal{E})$.

Furthermore, if (X, τ) is a topology and there exists a collection $\mathcal{E}$ of subsets of X such that $\tau(\mathcal{E}) = \tau$, then we say that $\mathcal{E}$ a *subbasis* of τ (or X).

Remark 1.1.6

Let X be a topological space and $\mathcal{E}$ be a subbasis of the topology on X . Then the definition of a subbasis gives that every element of $\mathcal{E}$ is open.

Using Definition 1.1.5, one can easily define a topology with desired open sets equipped. However, there exists some special subbasis which gives not only generates a topology, but also gives an analogous of Exercise 1.1.2.

Definition 1.1.7 (Basis and Local Basis)

Let (X, τ) be a topological space, and consider a collection $\mathcal{B}$ of some subsets of X such that

- (a) For each $x \in X$, there exists $V \in \mathcal{B}$ such that $V \ni x$;
- (b) If $U, V \in \mathcal{B}$ and $x \in U \cap V$, there exists $W \in \mathcal{B}$ such that $x \in W$ and $W \subset U \cap V$.

Then $\mathcal{B}$ is called a *basis* for τ if $\tau = \tau(\mathcal{B})$.

A *local basis* for τ at $x \in X$ is a collection $\mathcal{B} \subset \tau$ such that

- (c) $x \in V$ for all $V \in \mathcal{B}$;
- (d) If V is an open set containing x , then there exists $W \in \mathcal{B}$ such that $W \subset V$.

Remark 1.1.8

Let X be a topological space and $\mathcal{B}$ be a basis of the topology on X . Then the definition of a basis gives that every element of $\mathcal{B}$ is open.

Here is an analogous of Exercise 1.1.2 if a topological space has a basis.

Proposition 1.1.9

Let X be a topological space and $\mathcal{B}$ be a basis of the topology on X . Then $U \subset X$ is open if and only if for every $x \in U$, there exists $V \in \mathcal{B}$ such that $x \in V \subset U$.

proof. Consider

$$\tau = \{U \subset X : \forall x \in U, \exists V \in \mathcal{B} \text{ such that } x \in V \subset U\}. \quad (1.1)$$

By Exercise 1.1.10, τ is a topology for X . The definition of τ clearly implies $\mathcal{B} \subset \tau$, hence we obtain $\tau(\mathcal{B}) \subset \tau$. This implies that every open set U is contained in τ , thus we proved the one side.

Conversely, fix any $U \in \tau$. Further assume that U is not empty, otherwise it is clear that U is open. For each $x \in U$ associates $V_x \in \mathcal{B}$ such that $x \in V_x \subset U$. Thus we have

$$U = \bigcup_{x \in U} V_x.$$

Since each V_x is open, the right hand side is also open, thus U is also open. Therefore, $U \in \tau(\mathcal{B})$, and thus $\tau \subset \tau(\mathcal{B})$. This proves the proposition. $\square$

Exercise 1.1.10

Prove that τ defined at eq. (1.1) is a topology for X .

If X is a topological space, we should define closures and interiors differently. However, if one studied well about closures and interiors in metric spaces, below definition makes sense.

Definition 1.1.11 (Closures and Interiors)

Let X be a topological space and A a subset of X . Then a *closure* of A is an intersection of all closed sets each of which contains A , and an *interior* of A is a union of all open sets each of which is contained in A .

We write a closure of A by $\overline{A}$, and an interior of A by A° .

Below exercise can be proved directly from the definition of closures and interiors. Thus in the entire of this book, we use the below exercise without mention it.

Exercise 1.1.12*

Prove the followings.

- (a) For every set A , $\overline{A}$ is closed, and A° is open.
- (b) $E = \overline{E}$ if and only if E is closed, and $U = U^\circ$ if and only if U is open.

There is a relationship between closures and interiors.

Proposition 1.1.13

Let X be a topological space and A a subset of X . Then $\overline{X \setminus A} = X \setminus A^\circ$, and $(X \setminus A)^\circ = X \setminus \overline{A}$.

proof. The second one follows from the first one. Indeed,

$$\overline{A} = \overline{X \setminus (X \setminus A)} = X \setminus (X \setminus A)^\circ.$$

By taking their complement, we obtain the second one from the first one.

To prove the first one, fix any closed set $F \supset X \setminus A$. Then we have $X \setminus F \subset A$. Since $X \setminus F$ is open, by the definition of an interior, $X \setminus F \subset A^\circ$, which implies $X \setminus A^\circ \subset F$. Since this holds for every closed set which contains $X \setminus A$, the definition of a closure implies $X \setminus A^\circ \subset \overline{X \setminus A}$.

On the other hand, clearly $A^\circ \subset A$ and A° is open. Hence, $X \setminus A \subset X \setminus A^\circ$, and $X \setminus A^\circ$ is closed. Therefore, by the definition of a closure, $\overline{X \setminus A} \subset X \setminus A^\circ$ is obtained. The proof is now finished. $\square$

Exercise 1.1.14*

Prove that $\overline{A \cap B} \subset \overline{A} \cap \overline{B}$ and $\overline{A \cup B} = \overline{A} \cup \overline{B}$. How about infinite union and intersection, respectively?

Now we define neighborhoods of a point in topological spaces. Although we defined this to be open in metric spaces, for some purpose, we define neighborhoods *need not to be open*.

Definition 1.1.15 (Neighborhoods)

Let X be a topological space and $x \in X$. Then a subset $N \subset X$ is called a *neighborhood* of x if its interior contains x . That is, $x \in N^\circ$. For each $x \in X$, define $\mathcal{N}_x$ a collection of all open neighborhoods of x .

Now we are going to study a closure more deeply. Using the definition of a closure, it is hard to show that whether a point lie in the closure or not. Below proposition explains the way to check such things.

Proposition 1.1.16

Let X be a topological space, and A a nonempty subset of X . Then $x \in \bar{A}$ if and only if for every open neighborhood V of x , V and A intersects.

proof. Fix $x \in \bar{A}$ and an open neighborhood V of x . If $V \cap A = \emptyset$, then $A \subset X \setminus V$. Since $X \setminus V$ is closed, by the definition of the closure, we must have $\bar{A} \subset X \setminus V$. However, in this case, we obtain

$$x \in \bar{A} \cap V = \emptyset,$$

which is a contradiction. Thus V and A should intersect.

Conversely, if $x \notin \bar{A}$, then $x \in X \setminus \bar{A} = (X \setminus A)^\circ$ because of Proposition 1.1.13. Observe that $(X \setminus A)^\circ$ is an open neighborhood of x which is disjoint with $\bar{A}$, hence with A . Therefore, we found an open neighborhood of x which is disjoint with A provided that $x \notin \bar{A}$. $\square$

1.2 Examples of Topologies

In this section, we study several examples in topology. First of all, if X is a topological space and Y is a nonempty subset of X , one can give a canonical topology on Y using the one of X .

Definition 1.2.1 (Subspace Topology)

Let X be a topological space with the topology τ , and $Y \subset X$ be nonempty set. Then one can consider the collection

$$\tau' := \{Y \cap V : V \in \tau\}.$$

Then (Y, τ') is a topological space. We say that Y equipped the *subspace topology* from X meaning that the topology on Y is τ' .

Exercise 1.2.2

Prove that the collection τ' in Definition 1.2.1 is a topology on Y .

Let X be a topological space and Y be a subspace of X . Definition 1.2.1 means that every open set in Y is the type $U \cap Y$ where U is open in X . How about closed sets? One can easily infer that there is an analogous one for closed sets.

Proposition 1.2.3

Let X be a topological space and Y a subspace of X . Then $E \subset Y$ is closed if and only if there exists a closed set F in X such that $E = F \cap Y$.

proof. If E is closed in Y , $Y \setminus E$ is open in Y . Thus by the definition of subspace topology, there exists an open set U in X such that $Y \setminus E = U \cap Y$. Then we have

$$\begin{aligned} E &= Y \setminus (Y \setminus E) = Y \setminus (U \cap Y) = Y \cap [X \setminus (U \cap Y)] \\ &= Y \cap [(X \setminus U) \cup (X \setminus Y)] = Y \cap (X \setminus U). \end{aligned}$$

Thus one side of the proposition is proved because $X \setminus U$ is closed in X .

On the other hand, fix $E \subset Y$ and assume that there exists a closed set F in X such that $E = F \cap Y$. Then consider $Y \setminus E = Y \setminus (F \cap Y)$. By the upper calculation, one can easily check that $Y \setminus (F \cap Y) =$

$Y \cap (X \setminus F)$. Since $X \setminus F$ is open in X , by the definition of the subspace topology, the set $Y \setminus E$ is open in Y , hence E is closed in Y . $\square$

Now consider that X is a totally ordered set with an order $\leq$. In this case, we can give a topology on X using its order.

Definition 1.2.4 (Order Space)

Let X be a totally ordered set with an order $\leq$. We equipped the topology on X generated by a subbasis consisting all rays $\{y : x < y\}$ and $\{z : z < x\}$ for every $x \in X$.

An *order space* is a totally ordered set equipped the topology described in above.

For instance, the standard topology for the real line $\mathbb{R}$ is the order topology, meaning that the standard topology for $\mathbb{R}$ is generated by the collection

$$\{(a, b) : -\infty \leq a \leq b \leq \infty\}.$$

If $\{X_\alpha\}_{\alpha \in A}$ is a collection of topological spaces, we have defined their product $X = \prod_{\alpha \in A} X_\alpha$ but yet we do not know what kind of topology it has naturally. For each $\alpha \in A$ consider the function $\pi_\alpha : X \rightarrow X_\alpha$ defined by $\pi_\alpha(x) = x(\alpha)$.

Definition 1.2.5 (Product topology)

Let A be a nonempty set and X_α are topological space for each $\alpha \in A$. We define a *product topology* on $X = \prod_{\alpha \in A} X_\alpha$ generated by a subbasis

$$\{\{\pi_\alpha \in V\} : \alpha \in A, V \text{ is open in } X_\alpha\}.$$

Recall that if $A = \{\alpha_j\}_{j=1}^n$ is finite, we identify $\prod_{j=1}^n X_{\alpha_j}$ and $X_{\alpha_1} \times \cdots \times X_{\alpha_n}$ with the canonical bijective map.

1.3 Continuity

We already studied continuous functions on calculus class or basic analysis class. We can also define continuity of functions on general topological spaces.

Definition 1.3.1 (Continuity)

Let X and Y be topological spaces. We say that a function $f : X \rightarrow Y$ is *continuous* if for every open set V in Y , the set $\{f \in V\}$ is open in X .

There is equivalent statements for continuity, which will used in common.

Theorem 1.3.2

Let X and Y be topological spaces, and $f : X \rightarrow Y$ be function. Then the followings are equivalent.

- (a) f is continuous.
 - (b) For every $A \subset X$, $f[\bar{A}] \subset \overline{f[A]}$.
 - (c) For every closed set F in Y , $\{f \in F\}$ is closed in X .
 - (d) For every point $x_0 \in X$, and an open neighborhood V of $f(x_0)$, there exists an open neighborhood W of x_0 such that $f(x) \in V$ whenever $x \in W$.
-

proof. [(i) $\Rightarrow$ (ii)] First of all, assume that f is continuous and fix any subset A of X . Fix $x \in \overline{A}$ and an open neighborhood V of $f(x)$. Since f is continuous, $\{f \in V\}$ is open in X and notice that it contains x . Thus there exists $a \in A \cap \{f \in V\}$ because of Proposition 1.1.16. Then we now have $f(a) \in f[A] \cap V$, and thus $f(x) \in \overline{f[A]}$ by using Proposition 1.1.16 once again. Since we take any element x in $\overline{A}$, $\overline{f[A]} \subset \overline{f[A]}$ is proved.

[(ii) $\Rightarrow$ (iii)] Fix any closed set E in Y . In order to prove that $\{f \in E\}$ is closed, we will prove that the closure of $\{f \in E\}$ coincide with itself.

Notice that if $y \in \overline{\{f \in E\}}$, there exists $x \in \{f \in E\}$ such that $y = f(x)$. Since $x \in \{f \in V\}$, $f(x) \in E$ holds. Thus we have $y = f(x) \in E$, hence $\overline{\{f \in E\}} \subset E$.

Since we assumed (ii), and E is closed, we find

$$f[\overline{\{f \in E\}}] \subset \overline{f[\{f \in E\}]} \subset \overline{E} = E. \quad (1.2)$$

Now we fix $x \in \overline{\{f \in E\}}$. Then by Equation (1.2), $f(x) \in f[\overline{\{f \in E\}}] \subset E$ holds, thus $x \in \{f \in E\}$.

This gives $\overline{\{f \in E\}} \subset \{f \in E\}$, proving that $\{f \in E\}$ is closed.

[(iii) $\Rightarrow$ (i)] Since $X \setminus \{f \in A\} = \{f \in Y \setminus A\}$, this part is clear.

[(i) $\Rightarrow$ (iv)] Fix a point $x_0 \in X$, and an open neighborhood V of $f(x_0)$. Since we assumed (i), the set $\{f \in V\}$ is open in X . Furthermore, clearly x_0 lies in that set. One can easily check that if we let $W = \{f \in V\}$, then $x \in W$ implies $f(x) \in V$.

[(iv) $\Rightarrow$ (i)] Fix an open set V of Y . Without losing generality, we assume that $\{f \in V\}$ is not empty. Fix an element $x_0 \in \{f \in V\}$. Then V is an open neighborhood of $f(x_0)$. As we assumed (iv), there exists an open neighborhood W of x_0 such that $f(x) \in V$ whenever $x \in W$. In other words, we find $x_0 \in W \subset \{f \in V\}$. Therefore, $\{f \in V\}$ is open because x_0 is arbitrary.

This proves the theorem. $\square$

Because of Theorem 1.3.2 (d), we can consider the following definition, which is a generalization of ϵ - δ argument on $\mathbb{R}^d$ (or on metric spaces).

Definition 1.3.3 (Continuity at Points)

Let X and Y be topological spaces. We say that a function $f : X \rightarrow Y$ is *continuous* at $x_0 \in X$ if for every open neighborhood V of $f(x_0)$, there exists an open neighborhood W of x_0 such that $f(x) \in V$ for every $x \in W$.

It is said that f is continuous on $E \subset X$ if f is continuous at every points in E .

If $Y = [-\infty, \infty]$, then there is a weaker version for continuity, which is called semicontinuity.

Definition 1.3.4 (Semicontinuity)

Let X be a topological space. We say that a function $f : X \rightarrow [-\infty, \infty]$ is *upper semicontinuous* at $x_0 \in X$ if for every real $y > f(x_0)$, there exists an open neighborhood V of x_0 such that $f(x) < y$ for every $x \in V$.

Similarly, we say that a function f is *lower semicontinuous* at $x_0 \in X$ if for every $y < f(x_0)$, there exists an open neighborhood V of x_0 such that $f(x) > y$ for every $x \in V$.

For a subset E of X , we say that f is upper semicontinuous on E if f is upper semicontinuous for every points in E ; we define similarly for lower semicontinuity.

We simply say USC for upper semicontinuous, and LSC for lower semicontinuous, respectively.

Now we are going to study properties of USC and LSC functions. We will focus on LSC specific. However, below propositions has an analogous one for USC functions because of Proposition 1.3.6.

Proposition 1.3.5

Let X be a topological space and f be $[-\infty, \infty]$ -valued function. Then f is LSC on X if and only if $\{f > y\}$ is open in X for every real y . In addition, f is USC on X if and only if $\{f < y\}$ is open in X for every real y .

proof. Assume that f is LSC, fix $y \in \mathbb{R}$, and $x \in \{f > y\}$. Notice that if there is no such x , then automatically $\{f > y\}$ is open, so we can assume that such x exists. Then the definition of upper semicontinuity gives an open neighborhood V of x such that $V \subset \{f > y\}$, proving that $\{f > y\}$ is open in X .

To prove the converse one, notice that f is clearly LSC at x where $f(x) = -\infty$. Thus it suffices to assume that $f(x) > -\infty$. In this case, one can take a real number y such that $y < f(x)$. By our assumption, $\{f > y\}$ is open in X which contains x , which shows that f is LSC at x .

One can obtain the result for USC by just following the proof in above by interchanging appropriate order relations. The proposition is now proved. $\square$

Proposition 1.3.6

Let $f, g : X \rightarrow [-\infty, \infty]$ be functions. Then

- (i) f is LSC if and only if $-f$ is USC.
- (ii) If both f and g are LSC, then $f \vee g$, $f \wedge g$, and $f + g$ are LSC, where the addition is assumed to be meaningful.
- (iii) f is continuous at $x \in X$ if and only if f is both USC and LSC at x .
- (iv) Let $(f_n)_n$ is a sequence of LSC functions such that $f_n \rightarrow f$ uniformly on X . Then f is also LSC.

proof. (i) Notice that for every real y , $\{f > y\} = \{-f < -y\}$. Thus Proposition 1.3.5 immediately gives the result.

(ii) Fix a real y . Then we have

$$\{f \vee g > y\} = \{f > y\} \cup \{g > y\}, \quad \{f \wedge g > y\} = \{f > y\} \cap \{g > y\}.$$

Together with Proposition 1.3.5, we proved that $f \vee g$ and $f \wedge g$ are both LSC provided that f and g are LSC.

To prove that $f + g$ is LSC, it is assumed that there is no pattern $\infty - \infty$, otherwise the addition is undefined. Observe that

$$\begin{aligned} \{f + g > y\} &= \{g > -\infty\} \cap \{f + g > y\} \\ &= \{g > -\infty\} \cap \{f > y - g\} \\ &= \{g > -\infty\} \cap \bigcup_{x \in \mathbb{R}} \{f > x\} \cap \{x > y - g\} \\ &= \left(\bigcup_{n=1}^{\infty} \{g > -n\} \right) \cap \bigcup_{x \in \mathbb{R}} \{f > x\} \cap \{g > y - x\}. \end{aligned}$$

Since f and g are LSC, the right most set is open by Proposition 1.3.5. Therefore, $\{f + g > y\}$ is also open, and thus $f + g$ is LSC.

- (iii) If f is both LSC and USC, by Proposition 1.3.5, both $\{f > a\}$ and $\{f < b\}$ are open in X for every real $a, b \in \mathbb{R}$, which implies $\{a < f < b\}$ is also open in X . Since the topology of $[-\infty, \infty]$ is generated by $[-\infty, a)$, (a, b) , and $(b, \infty]$, this proves that f is continuous. The converse one is obvious thanks to Proposition 1.3.5.
- (iv) Fix $y \in \mathbb{R}$, and $x_0 \in \{f > y\}$. Remember that we can assume that such x_0 exists, otherwise, $\{f > y\}$ will be empty and thus clearly open. Fix $\epsilon > 0$ will be specified later. Since $f_n \rightarrow f$ uniformly on X as $n \rightarrow \infty$, there exists a positive integer N such that $|f_n - f| < \epsilon$ on X for every $n \geq N$. Especially, we have $f_n(x_0) > f(x_0) - \epsilon$. As each f_n is LSC, for each n there exists a neighborhood V_n of x_0 such that $f_n(x) > f(x_0) - \epsilon$. Notice that for every $n \geq N$, we have $f_n(x) > f_N(x) - \epsilon$ for every $x \in X$. Therefore, on $V = V_1 \cap \dots \cap V_N$, we have

$$f_n(x) > f(x_0) - 2\epsilon, \quad \forall x \in V.$$

This implies that $f(x) > f_n(x) - \epsilon > f(x_0) - 3\epsilon$ for every $x \in V$. Now take $3\epsilon = f(x_0) - y$ and by observe that V is open. Hence we find an open set V such that $x_0 \in V \subset \{f > y\}$, which proves that f is LSC on X .

The proposition is now proved. □

1.4 Separation Axioms

In the topological space, we consider the “grade” in the sense that open sets can “separate” points well.

Definition 1.4.1 (Separation Axioms)

Let X be a topological space. We say that X is

- (i) T_0 space if $x, y \in X$ there exists an open set U such that either $x \in U$ and $y \notin U$, or $y \in U$ and $x \notin U$.
- (ii) T_1 space if $x, y \in X$ there exist open sets $U \in \mathcal{N}_x$ and $V \in \mathcal{N}_y$ such that $y \notin U$ and $x \notin V$.
- (iii) T_2 space if $x, y \in X$ there exist *disjoint* open sets $U \in \mathcal{N}_x$ and $V \in \mathcal{N}_y$.
- (iv) T_3 space if $x \in X$ and a closed set C in X there exist *disjoint* open sets U and V such that $x \in U$ and $C \subset V$.
- (v) T_4 space if C and D are closed sets in X there exist *disjoint* open sets U and V such that $C \subset U$ and $D \subset V$.
- (vi) T_5 space if C and D are *separated* sets in X there exist *disjoint* open sets U and V such that $C \subset U$ and $D \subset V$. Here, we say that two sets C and D are separated if $\overline{C} \cap D = C \cap \overline{D} = \emptyset$.
- (vii) T_6 space if C and D are closed sets in X there exists a continuous function $f : X \rightarrow [0, 1]$ such that $\{f = 0\} = C$ and $\{f = 1\} = D$.

We also called

- (a) T_2 space is also called by *Hausdorff space*.
- (b) A *regular space* is both T_1 and T_3 spaces.
- (c) A *normal space* is both T_1 and T_4 spaces.

- (d) A *completely normal space* is both T_1 and T_5 spaces.
- (e) A *perfectly normal space* is both T_1 and T_6 spaces.

1.4.1 T_1 spaces

First of all, we will study a proposition related to T_1 spaces.

Proposition 1.4.2

Assume that X contains at least two points. Then the space X is T_1 if and only if every singleton is closed.

proof. Fix $x \in X$. Then for each $y \in X$ distinct with x , there exists an open set U such that $x \notin U$ and $y \in U$. In other words, we find an open set U satisfies $y \in U \subset X \setminus \{x\}$. As y is an arbitrary points of $X \setminus \{x\}$, the set is open, hence $\{x\}$ is closed.

Conversely, assume that every singleton is closed. Fix two distinct points x and y in X . Then take $U = X \setminus \{y\}$ and $V = X \setminus \{x\}$ satisfies the property in the definition of T_1 in Definition 1.4.1. The proposition is now proved. $\square$

Exercise 1.4.3

From the proof of Proposition 1.4.2, one might think that every singleton should be closed also in T_0 spaces. Prove or disprove that on T_0 spaces, every singleton is closed.

1.4.2 Hausdorff spaces

Now we are going to study some properties of Hausdorff spaces. First theorem gives an equivalent condition whether a space is Hausdorff.

Theorem 1.4.4

The space X is Hausdorff if and only if the diagonal set

$$\Delta := \{(x, x) : x \in X\}$$

is closed in the product space $X \times X$.

Exercise 1.4.5

Let X be a topological space and Y be Hausdorff space. Let f and g are continuous function from X into Y . Prove that the set

$$\{f = g\} := \{x \in X : f(x) = g(x)\}$$

is closed in X . What if the condition “Hausdorff” is dropped for Y ?

Every Hausdorff space is indeed T_1 . Here is the proof.

Proposition 1.4.6

Every Hausdorff space is T_1 .

proof. Let X be any Hausdorff space. There is nothing to prove if X is singleton, thus we assume that X contains at least two points. Fix $x_0 \in X$. Then for any $y \in X$, we can take two disjoint

open sets $U \in \mathcal{N}_{x_0}$ and $V \in \mathcal{N}_y$. In other words, we have $y \in V \subset X \setminus \{x_0\}$. Since y is arbitrary element of $X \setminus \{x_0\}$, this proves that $X \setminus \{x_0\}$ is open, hence $\{x_0\}$ is closed. As x_0 is also choen arbitrary in X , this proves that X is T_1 because of Proposition 1.4.2. $\square$

1.4.3 Regular spaces and normal spaces

Theorem 1.4.7 (Urysohn's Lemma)

Let X be a normal space. If C and D are disjoint closed sets, there exists a continuous function $\phi : X \rightarrow [0, 1]$ such that $\phi \equiv 0$ on C , and $\phi \equiv 1$ on D .

1.4.4 Completely normal spaces

Theorem 1.4.8

The space X is completely normal if and only if every subspace of X is normal.

proof. First, assume that X is completely normal, fix any subspace Y of X , and fix two disjoint closed subsets A and B of Y . Without loss of generality, assume that both A and B are not empty. Take two closed sets E and F in X such that

$$A = E \cap Y, \quad B = F \cap Y$$

by Proposition 1.2.3. Since A and B are not empty, so are E and F , respectively. Furthermore, as A and B are disjoint, E and F are also disjoint. From this fact and Exercise 1.1.14, we have

$$\overline{A} \cap B = \overline{E \cap Y} \cap (F \cap Y) \subset (\overline{E} \cap F) \cap (\overline{Y} \cap Y) = E \cap F \cap Y = \emptyset,$$

showing that $\overline{A} \cap B$ is empty. Similarly, one can obtain that $A \cap \overline{B}$ is empty, thus A and B are separated. Since X is completely normal, there exist disjoint open sets U and V such that $U \supset A$ and $V \supset B$. Then $U \cap Y$ and $V \cap Y$ are open in Y , disjoint, and contains A and B , respectively. Therefore, we proved that Y is normal because A and B can be any disjoint closed sets.

[Fill Here!]

Remark 1.4.9

If the space X is just normal, there is no guarantee that a subspace is also a normal space. See [7] for the counterexample.

The example of completely normal space is an order space. One can see its proof in [7]. Later (see Remark 2.3.11) we use this fact to give some counterexample.

1.4.5 Perfectly normal spaces

Now we are going to study perfectly normal spaces. Like completely normal spaces, there is an equivalent statement with the definition of perfectly normal spaces.

Theorem 1.4.10 (Vedenissov's Theorem)

Let X be a topological space. Then the following statements are equivalent.

- (i) X is perfectly normal.
- (ii) X is T_1 and for every open set V , there exists a continuous function $\phi : X \rightarrow [0, 1]$ such that $V = \{\phi > 0\}$.
- (iii) X is normal, and every closed set is G_δ , that is, every closed set is a countable intersection of open sets.

proof. If X is perfectly normal, by definition, it is T_1 space. Fix an open set V . If V is empty, just take $\phi \equiv 0$. Otherwise, fix any point $x_0 \in V$. By Proposition 1.4.2, the set $\{x_0\}$ is closed. By the definition of T_6 spaces, there exists a continuous function $\phi : X \rightarrow [0, 1]$ such that $\{\phi = 0\} = X \setminus V$, and $\{\phi = 1\} = \{x_0\}$. This, therefore, implies $V = \{\phi > 0\}$.

On the other hand, if the converse assumption is satisfied, fix two disjoint closed sets C and D . Take two continuous functions $\phi : X \rightarrow [0, 1]$ and $\psi : X \rightarrow [0, 1]$ such that

$$X \setminus C = \{\phi > 0\}, \quad X \setminus D = \{\psi > 0\}.$$

Since C and D are disjoint, we have

$$C \subset X \setminus D = \{\psi > 0\}, \quad D \subset X \setminus C = \{\phi > 0\}.$$

Now define

$$\theta(x) = \frac{\phi(x)}{\phi(x) + \psi(x)}, \quad \forall x \in X.$$

Then one can easily see that $0 \leq \theta \leq 1$, θ is continuous, $\{\theta = 0\} = \{\phi = 0\} = C$, and $\{\theta = 1\} = \{\psi = 0\} = D$. Thus we proved that (i) and (ii) are equivalent. For that θ , we can also prove that if X is perfectly normal, then X is normal. Indeed, consider $U = \{\theta < 1/2\}$, and $V = \{\theta > 1/2\}$. Since $C = \{\theta = 0\}$ and $D = \{\theta = 1\}$, $C \subset U$, and $D \subset V$. Furthermore, it is clear that U and V are disjoint, hence X is normal as we can take arbitrary such C and D .

Now fix a closed set F in X . From (ii), there exists a continuous function $\phi : X \rightarrow [0, 1]$ such that $X \setminus F = \{\phi > 0\}$, that is,

$$F = \{\phi = 0\} = \bigcap_{n=1}^{\infty} \left\{ \phi < \frac{1}{n} \right\}.$$

Now notice that $\{\phi < 1/n\}$ is open in X for every $n = 1, 2, 3, \dots$, because ϕ is continuous.

On the other hand, assume that (iii) holds. Since X is normal, we can apply Theorem 1.4.7. Fix any closed set F in X . Since F is G_δ , one can take a sequence of open sets $(V_n)_n$ such that $F = \bigcap_{n=1}^{\infty} V_n$. Then for each n , use Theorem 1.4.7 to take a continuous function $\phi_n : X \rightarrow [0, 1]$ such that $\phi_n \equiv 0$ on F , and $\phi_n \equiv 1$ on $X \setminus V_n$. Then the function

$$\phi = \sum_{n=1}^{\infty} \frac{\phi_n}{2^n}$$

is the desired one to make $F = \{\phi = 0\}$. Therefore, we proved that (ii) and (iii) are equivalent. The proof is now finished. $\square$

Remark 1.4.11

Theorem 1.4.7 implies that for any disjoint closed sets C and D , there exists a continuous function $\phi : X \rightarrow [0, 1]$ such that $C \subset \{\phi = 0\}$, and $D \subset \{\phi = 1\}$. From this perspective, perfect normality

is much stronger than normality.

To end this section, we will prove relations between several separation axioms.

1.5 Nets

In the basis analysis, one is taught that if X is a metric space and A is a subset of X then $x \in \bar{A}$ if and only if there is a sequence $(x_n)_n$ in A that converges to x . However, such fact does not hold for general topological spaces. Actually, several properties of sequences in metric spaces break when we generalize those in the arbitrary topological spaces. This makes a reason introducing an object which acts like a sequence more general sense. There are two things that generalize sequences: nets, and filters. In here, we are going to study about nets. Filter theory is introduced in the appendix.

First, we need to see what are nets. Note that nets are defined even if X has no topology.

Definition 1.5.1 (Nets)

Let X be a nonempty set. A *directed set* is a nonempty set A equipped with a preorder $\leq$ (reflexive and transitive relation) satisfying the following condition:

- (i) For $\alpha, \beta \in A$, there exists $\gamma \in A$ such that $\alpha \leq \gamma$ and $\beta \leq \gamma$.

A *net* in X is a function $x : A \rightarrow X$ from a directed set A to X . We usually write $(x_\alpha)_{\alpha \in A}$ or simply (x_α) .

Below definition will be used to define a convergence of net later.

Definition 1.5.2 (Eventually and Frequently)

We say that a net $(x_\alpha)_{\alpha \in A}$ is *eventually* in $E \subset X$ if there exists $\alpha_0 \in A$ such that $x_\alpha \in E$ for all $\alpha \geq \alpha_0$.

A net $(x_\alpha)_{\alpha \in A}$ is *frequently* in $E \subset X$ if for every $\alpha \in A$ there exists $\beta \geq \alpha$ such that $x_\beta \in E$.

Remark 1.5.3

Every sequence is a net because clearly $\mathbb{N}$ is a directed set with an usual ordering. From the definition, the term “eventually” implies “frequently” but not the converse. For instance, consider a net (actually a sequence) $((-1)^n)_{n \in \mathbb{N}}$. This net does not eventually in the interval $(0, 2)$ and this fact is easily obtained from the definition. However, for every n , take any even number m greater than n . In this case, since $(-1)^m = 1 \in (0, 2)$, given net is frequently in $(0, 2)$.

Since nets is a generalization of sequences, it is natural to consider what is the analogous definition of subsequences for nets. The definition of which is the following.

Definition 1.5.4 (Subnet)

Let X be a nonempty set. A net $(y_\beta)_{\beta \in B}$ is a *subnet* of a net $(x_\alpha)_{\alpha \in A}$ if there exists a function $h : B \rightarrow A$ such that

- (i) For every $\alpha \in A$, there exists $\beta_0 \in B$ such that $h(\beta) \geq \alpha$ for all $\beta \geq \beta_0$.
- (ii) $y_\beta = x_{h(\beta)}$.

Remark 1.5.5

Clearly every net is a subnet of itself. Unlike sequences, the cardinality of B might be larger than A .

There is an important relation between terminologies “eventually” and “frequently”.

Proposition 1.5.6

Let X be a nonempty set, $(x_\alpha)_{\alpha \in A}$ be a net which is frequently in some subset $E \subset X$. Then there exists a subnet which is eventually in E .

proof.

1.6 Compact sets

In this section, we will learn about compact sets in arbitrary topological spaces.

Definition 1.6.1 (Compact sets)

Let X be a topological space. Then a set X is said to be *compact* if for every collection $\mathcal{U}$ of open sets which cover X , we can find a finite subcollection $\mathcal{F} \subset \mathcal{U}$ such that $\bigcup \mathcal{F} \supset X$. The collection of type $\mathcal{U}$ described in above is called an *open cover of X* . A subset $K \subset X$ is said to be compact if K is compact within the subspace topology. A set is said to be *precompact* if the closure of which is compact.

We can show that the space is compact by using nets. Here is the theorem about.

Theorem 1.6.2

Let X be a topological space. Then X is compact if and only if every universal net in X converges.

proof. Suppose that there is a universal net $(x_\alpha)_{\alpha \in A}$ which converges nowhere in X . For each $x \in X$ take an open set U_x which the net is not eventually in. Note that clearly $\{U_x\}_{x \in X}$ is an open cover of X . If X were compact, there are finitely many points $x_1, \dots, x_n$ such that

$$U_{x_1} \cup U_{x_2} \cup \dots \cup U_{x_n} \supset X. \quad (1.3)$$

However as $(x_\alpha)_{\alpha \in A}$ is a universal net, it is eventually in $X \setminus U_x$ for each $x \in X$ by ???. This and eq. (1.3) gives

$$(x_\alpha)_{\alpha \in A} \text{ is eventually in } (X \setminus U_{x_1}) \cap (X \setminus U_{x_2}) \cap \dots \cap (X \setminus U_{x_n}) = \emptyset,$$

which is a contradiction. Such contradiction occurs as we assumed that X is compact. Hence X cannot be compact.

Conversely, suppose that X is not compact. Then there is an open cover $\mathcal{U}$ which does not have a finite subset of $\mathcal{U}$ that covers X . Put

$$\mathcal{V} := \{X \setminus E : E \in \mathcal{U}\}.$$

Observe that $\mathcal{V}$ satisfies FIP. Take a universal net $(x_\alpha)_{\alpha \in A}$ which is eventually in every set in $\mathcal{V}$ by ??? and ???. Since $\mathcal{U}$ is an open cover of X , there is no $x \in X$ such that $x_\alpha \rightarrow x$. Indeed, if $x \in X$ then there is $U \in \mathcal{U}$ that contains x . Since $X \setminus U \in \mathcal{V}$, the net is eventually in $X \setminus U$ and this implies that the net does not converge to x . $\square$

Corollary 1.6.3

Let X be a topological space. Then nonempty subset $K \subset X$ is compact if and only if every universal net in X eventually in K converges to some point in K . Notice that the convergent point can differ.

proof. Suppose that $K \subset X$ is compact, and fix a universal net $(x_\alpha)_{\alpha \in A}$ that is eventually in K . Then the subnet $(y_\beta)_{\beta \in B}$ restricted to indices where $x_\alpha \in K$ is a universal net in K by Exercise 1.6.4. Being compact, $y_\beta \rightarrow x \in K$ by Theorem 1.6.2. Now fix an open set N which contains x . Since $N \cap K$ is open in K and $x \in N \cap K$ we have $(y_\beta)_{\beta \in B}$ is eventually in $N \cap K$ because it converges to x in K .

In order to prove that $(x_\alpha)_{\alpha \in A}$ is eventually in N , assume that it is not. Then by ??, it is eventually in $X \setminus N$, thus $(y_\beta)_{\beta \in B}$ is eventually in $(X \setminus N) \cap K$. Since it is also eventually in $N \cap K$, we must obtain

$$(y_\beta)_{\beta \in B} \text{ is eventually in } (N \cap K) \cap [(X \setminus N) \cap K] = \emptyset,$$

which is a contradiction. Therefore, $(x_\alpha)_{\alpha \in A}$ is eventually in N , and thus we prove that $x_\alpha \rightarrow x$ since N is arbitrary.

To prove the sufficiency, claim that if $(y_\beta)_{\beta \in B}$ is any universal net in K , then there exists a universal net $(x_\alpha)_{\alpha \in A}$ in X eventually in K whose subnet corresponds to it. By our assumption, $(x_\alpha)_{\alpha \in A}$ converges to some $x \in K$. By the definition of the subspace topology, it is easy to verify that $y_\beta \rightarrow x$. Therefore, K is compact because of Theorem 1.6.2. $\square$

Exercise 1.6.4*

Prove that the restricted subnet in K in Corollary 1.6.3 is really a universal net in K .

Below propositions can be proved without using any net theory. In here, however, we will see how Theorem 1.6.2 (and Corollary 1.6.3) can be used to prove several compactness properties. One may solve Exercise 1.6.6.

Proposition 1.6.5

Let X be a topological space and $K \subset X$ be a nonempty set.

- (a) If X is Hausdorff and K is compact, then K is closed.
- (b) If X is compact and K is closed, then K is compact.
- (c) Let Y be another topological space and $f : X \rightarrow Y$ a continuous map, then $f[K]$ is compact in Y whenever K is compact in X .

proof. (a) Fix $x \in \overline{K}$ and take a subnet $(y_\beta)_{\beta \in B}$ in X such that it is in K and $y_\beta \rightarrow x$ by ??. Then take a universal subnet $(z_\gamma)_{\gamma \in C}$ by ??. Since $y_\beta \rightarrow x$ and $(z_\gamma)_{\gamma \in C}$ is a subnet, we have $z_\gamma \rightarrow x$. On the other hand, since K is compact, and $(z_\gamma)_{\gamma \in C}$ is eventually in K , by Corollary 1.6.3, there exists $y \in K$ such that $z_\gamma \rightarrow y$. Since X is Hausdorff, x and y must be equal because of ??, proving that $x = y \in K$. Since x is arbitrary element of $\overline{K}$, we proved that K is closed.

(b) Fix a universal net $(x_\alpha)_{\alpha \in A}$ in X which is eventually in K . Being compact, $(x_\alpha)_{\alpha \in A}$ converges to some point $x \in X$. By ??, $x \in \overline{K} = K$. Hence, K is compact because of Corollary 1.6.3 as $(x_\alpha)_{\alpha \in A}$ is arbitrary.

(c) Fix a universal net $(z_\gamma)_{\gamma \in C}$ in $f[K]$. Then take a universal net $(x_\alpha)_{\alpha \in A}$ in K such that $f_*(x_\alpha)$ is a subnet of $(z_\gamma)_{\gamma \in C}$. Being compact, we can take a point $x \in K$ such that $x_\alpha \rightarrow x$ by Theorem 1.6.2. Since f is continuous, $f_*(x_\alpha) \rightarrow f(x)$ because of ??. Observe that since both are universal nets and one is a subnet of another, they converge to the same points. Therefore, $z_\gamma \rightarrow f(x)$, and we proved that $f[K]$ is compact by Theorem 1.6.2 as $(z_\gamma)_{\gamma \in C}$ is arbitrary. $\square$

Exercise 1.6.6

Prove Proposition 1.6.5 without using any net theories.

We studied what is a compact set, how this is related to filter theory, and studied some properties about compact sets. Checking whether a set is compact is very difficult. On the ordered space, under some conditions, every closed interval is compact.

Theorem 1.6.7

Let X be an order space which has a least upper bound property. Then every closed interval is compact.

proof. Fix any closed interval $[a, b]$, and an ultrafilter $\mathcal{U}$ in X which contains $[a, b]$. Consider

$$A := \{x \in [a, b] : [x, b] \in \mathcal{U}\}.$$

Since $[a, b] \in \mathcal{U}$, the set A is not empty, and clearly A has an upper bound, which is b . Thus the supremum of A exists, let us call it α . Claim that $\mathcal{U} \rightarrow \alpha$. Since filters are closed under finite intersection and contains all supersets of their elements, it suffices to consider open rays in $[a, b]$ which contain α .

First, consider $(x, b]$ where $x \in [a, b]$ and that $\alpha \in (x, b]$. Since α is the supremum and $x < \alpha$, there exists $\beta \in (x, \alpha] \cap A$. That means, $[\beta, b] \in \mathcal{U}$. Since $(x, b] \supset [\beta, b]$, we thus obtain $(x, b] \in \mathcal{U}$. Now consider $[a, x)$ where $x \in (a, b]$ and that $\alpha \in [a, x)$. If $[a, x)$ does not belong to $\mathcal{U}$, Proposition A.1.5 and $[a, b] \in \mathcal{U}$ imply

$$[x, b] = (X \setminus [a, x)) \cap [a, b] \in \mathcal{U},$$

thus $x \in A$. However we take α the supremum of A and we shows that x belongs to A although $x > \alpha$, which is a contradiction indeed. Therefore, $[a, x) \in \mathcal{U}$ should be true, and we proved that $\mathcal{U} \rightarrow \alpha$.

Since $\mathcal{U}$ be any ultrafilter in X containing $[a, b]$, ?? gives that $[a, b]$ is compact. $\square$

Compact spaces are also related to several separation axioms. One of the most important theorem is the following.

Theorem 1.6.8

Every compact Hausdorff space is normal.

proof. Let X be any compact Hausdorff space. Fix any two disjoint closed subsets A and B of X . Being compact, both A and B are compact by Proposition 1.6.5. Fix $a \in A$. Since X is Hausdorff, for each $b \in B$, we can take a disjoint open sets $U_{ab} \in \mathcal{N}_a$ and $V_{ab} \in \mathcal{N}_b$. Then we have an open cover $\{V_{ab}\}_{b \in B}$ of B , thus there exists finitely many points $b_1, \dots, b_n \in B$ such that

$$B \subset V_{ab_1} \cup \dots \cup V_{ab_n}$$

because B is compact. We now define

$$U_a := \bigcap_{j=1}^n U_{ab_j}, \quad V_a := \bigcup_{j=1}^n V_{ab_j}.$$

Then U_a is an open set containing a , $V_a \supset B$, and U_a and V_a are disjoint. Notice that such U_a and V_a can be obtained for every $a \in A$.

Now we have an open cover $\{U_a\}_{a \in A}$ of A . Being compact, there exists a finitely many points $a_1, \dots, a_m \in A$ such that

$$A \subset U_{a_1} \cup \dots \cup U_{a_m}.$$

Now two open sets

$$U := \bigcup_{j=1}^m U_{a_j}, \quad V := \bigcap_{j=1}^m V_{a_j}$$

contains A and B , respectively, and they are disjoint. Therefore we proved that the space X is T_4 . Since X is Hausdorff, it is T_1 by Proposition 1.4.6, hence we proved that X is normal. $\square$

On a compact space, semicontinuity gives a uniform convergence when the sequence is increasing. Such is so called Dini's convergence theorem. Although the original one just say about continuous function, we can generalized into semicontinuous functions.

Theorem 1.6.9 (Dini's Convergence Theorem)

Let X be a compact topological space, and $(f_n)_n$ be a sequence of real-valued LSC functions such that $f_n \uparrow f$ as $n \rightarrow \infty$ where f is continuous. In other words, $f_n \leq f_{n+1} \leq f$ for every n and $f_n \rightarrow f$ as $n \rightarrow \infty$. Then $f_n \rightarrow f$ uniformly on X as $n \rightarrow \infty$.

proof. By considering $g_n = f_n - f$, which is still LSC because of Proposition 1.3.6, we can assume that $f = 0$ and $f_n \leq 0$ for every n .

Fix $\epsilon > 0$ and consider sets $K_n = \{f_n \leq -\epsilon\}$. Since f_n is LSC, each K_n is closed by Proposition 1.3.5, hence compact since X is compact. Notice that $K_n \supset K_{n+1}$ for every n , and

$$\bigcap_{n=1}^{\infty} K_n = \{0 \leq -\epsilon\} = \emptyset.$$

Hence, by Proposition 1.6.5, there must exist a positive integer N such that K_N is empty. This means that for every $n \geq N$, $-\epsilon < f_n \leq 0$ on X , therefore we proved that $f_n \rightarrow 0$ uniformly on X . $\square$

Remark 1.6.10

On the proof of Theorem 1.6.9, we just use the upper semicontinuity of f . However, as we proved the uniform convergence, f will be LSC, hence f should be continuous.

We close this section by proving remarkable theorem, so called Tychonoff's theorem. This states that product of compact spaces remains compact under the product topology.

Theorem 1.6.11 (Tychonoff's Theorem)

Let $\{X_\alpha\}_{\alpha \in A}$ be a collection of compact topological spaces where A is an index set. Then the space $X = \prod_{\alpha \in A} X_\alpha$ equipped with the product topology is compact.

proof. Let us define a projection $\pi_\alpha : X \rightarrow X_\alpha$ for each $\alpha \in A$, and fix an ultrafilter $\mathcal{U}$ in X . By Proposition A.1.15, $(\pi_\alpha)_* \mathcal{U}$ is an ultrafilter in X_α , so we can take a point $x_\alpha \in X_\alpha$ such that $(\pi_\alpha)_* \mathcal{U} \rightarrow x_\alpha$. Such points exists because X_α is compact and recall ???. Finally, take the point $x \in X$ such that $\pi_\alpha(x) = x_\alpha$ for every $\alpha \in A$. Claim that $\mathcal{U} \rightarrow x$.

To prove this, fix an open neighborhood N of x in X . Since

$$\{\{\pi_\alpha \in U\} : \alpha \in A, U \in X_\alpha\}$$

forms a subbasis of X , there exists $\alpha_1, \dots, \alpha_n \in A$, and $U_{\alpha_1}, \dots, U_{\alpha_n}$ are open sets in X_{α_j} , $j = 1, 2, \dots, n$, respectively, such that

$$x \in \{\pi_{\alpha_1} \in U_{\alpha_1}\} \cap \{\pi_{\alpha_2} \in U_{\alpha_2}\} \cap \dots \cap \{\pi_{\alpha_n} \in U_{\alpha_n}\} \subset N.$$

Notice that $x_{\alpha_j} \in U_{\alpha_j}$ for $j = 1, 2, \dots, n$. This gives that $U_{\alpha_j} \in (\pi_{\alpha})_* \mathcal{U}$, hence

$$\{\pi_{\alpha_j} \in U_{\alpha_j}\} \in \pi_{\alpha}^*(\pi_{\alpha})_* \mathcal{U} \subset \mathcal{U}.$$

Thus,

$$\{\pi_{\alpha_1} \in U_{\alpha_1}\} \cap \{\pi_{\alpha_2} \in U_{\alpha_2}\} \cap \dots \cap \{\pi_{\alpha_n} \in U_{\alpha_n}\} \in \mathcal{U},$$

and this gives $N \in \mathcal{U}$ because $\mathcal{U}$ is an ultrafilter. Since N is arbitrary open neighborhood of x , we have proved that $\mathcal{U} \rightarrow x$. Therefore, X is compact because of ?? □

Remark 1.6.12

To prove Theorem 1.6.11, we need the axiom of choice. However at the first glance, there is no axiom of choice is used. But remember that while proving Proposition A.1.2 we use the axiom of choice. In addition, the space X_{α} might not be Hausdorff. This means that one must “choose” one x_{α} from several points at which the ultrafilter $(\pi_{\alpha})_* \mathcal{U}$ converges.

Actually, Theorem 1.6.11 and the axiom of choice are equivalent. This means that one cannot prove Theorem 1.6.11 without using the axiom of choice although one can use the ultrafilter lemma (see Remark A.1.4). However, if all spaces X_{α} are *Hausdorff*, then Theorem 1.6.11 can be proved only applying the ultrafilter lemma not using the axiom of choice.

Remark 1.6.13

If the index set A is finite, we can prove that the product space is compact without using the axiom of choice. Notice that in this case, we cannot apply any filter theory, especially ??, since this uses ultrafilter lemma, and we proved this using the axiom of choice (see Proposition A.1.2 and Remark A.1.4). One can see the proof the finite case at [1, 6] for instance.

1.7 Hints for Exercises

[Exercise 1.4.3] Find a topological space which is T_0 but not T_1 .

Measures and Integrations

2.1 Introduction

What is the definition of the “area” of some figures? In the *real* kindergarten, the reader is taught about the area of a rectangle is calculated by a multiple of the width and the height of the rectangle. In the high school, the reader learn how to calculate the circle by approximating areas of rectangles which fill the whole circle from inside. Generally speaking, one can infer that every figure has an area and it can be calculated with similar mannar.

In our intuition, a length —an area in $\mathbb{R}$ — is translation invariant, and it should be in our real life. Let us denote ℓ a length of sets in $\mathbb{R}$ and assume that we can take a length from arbitrary sets in $\mathbb{R}$. For instance, $\ell([a, b]) = b - a$ and such definition comes from our intuition.

The Entire settings of this chapter is heavily inspired from [5] and [4]. If one already knows measure theory, read Remark 2.2.7 first, and then read whole chapter because this textbook is written in different manner.

2.2 Measures and measurable sets

In this section, we will learn about measures and their properties. First two definitions introduce what is a measure and what is a measurable set.

Definition 2.2.1 (Measures and measure spaces)

Let X be a nonempty set. A *measure* μ on X is a function $\mu : \mathcal{P}(X) \rightarrow [0, \infty]$ such that

- (i) $\mu(\emptyset) = 0$;
- (ii) If $A \subset B$, then $\mu(A) \leq \mu(B)$.
- (iii) $\mu(\bigcup_{j=1}^{\infty} A_j) \leq \sum_{j=1}^{\infty} \mu(A_j)$.

We say the pair (X, μ) a *measure space*.

Below exercise is almost trivial, so we use these facts without mention it. However, we should prove the following property because it is not implied from the definition directly.

Exercise 2.2.2

Let X be a nonempty set and μ a measure on X . Then for every positive integer n ,

$$\mu\left(\bigcup_{j=1}^n A_j\right) \leq \sum_{j=1}^n \mu(A_j).$$

Now we are going to define a term “measurable” of sets.

Definition 2.2.3 (Measurable sets)

Let (X, μ) be a measure space. A set $A \subset X$ is called μ -measurable if for every set $E \subset X$, the following relation holds.

$$\mu(E) = \mu(E \cap A) + \mu(E \setminus A).$$

Define $\mathfrak{M}_\mu$ the collection of all μ -measurable sets.

Definition 2.2.4 (Null set)

Let (X, μ) be a measure space. A subset $E \subset X$ is called a μ -null set or just null set in short if $\mu(E) = 0$.

Exercise 2.2.5*

On a measure space (X, μ) , A set $A \subset X$ is μ -measurable if and only if for every $E \subset X$ with $\mu(E) < \infty$,

$$\mu(E) \geq \mu(E \cap A) + \mu(E \setminus A).$$

Exercise 2.2.6*

Prove that every μ -null set is μ -measurable.

Remark 2.2.7

Actually, the object μ defined in Definition 2.2.1 is so called an *outer measure* in several references, and reserve the term *measure* for different manner. Furthermore, the term *measure space* stands for a triple $(X, \mathfrak{M}, \mu)$ where $\mathfrak{M}$ is so-called σ -field (definition of which is introduced in Definition 2.2.8) and μ a measure (not an outer measure). Although there is some confusion while reading this textbook, it has some benefits when we define terms *measure* and *measure space* by Definition 2.2.1. So reader should beware from misunderstanding.

So far we have defined a measure μ and the collection $\mathfrak{M}_\mu$ of μ -measurable sets on a set X . Below definition gives a term which $\mathfrak{M}_\mu$ will satisfy, and the theorem proves it.

Definition 2.2.8

Let X be a nonempty set. A subcollection $\mathfrak{M} \subset \mathcal{P}(X)$ is called a σ -field if

- (i) $\emptyset, X \in \mathfrak{M}$.
- (ii) If $A \in \mathfrak{M}$, then $X \setminus A \in \mathfrak{M}$.
- (iii) If $\{A_n\}_{n=1}^\infty \subset \mathfrak{M}$ is a countable subcollection, then $\bigcup_{n=1}^\infty A_n \in \mathfrak{M}$. Notice that $\{A_n\}_n$ need not to be mutually disjoint.

If the condition *countable* is replaced by *finite* in (iii), then the collection $\mathfrak{M}$ is called by a *field*.

Remark 2.2.9

Terms *field* and σ -*field* are usually called by probabilists. In other people who study analysis, *algebra* and σ -*algebra* is used, respectively.

However, personally, the former terms is the *right* term in the sense that *fields* (resp. σ -*fields*) are closed under all set theoretical operations $\cup$, $\cap$, $\setminus$, and the complement $\cdot^c$.

Exercise 2.2.10*

Let X be a nonempty set. A subcollection $\mathfrak{M} \subset \mathcal{P}(X)$ satisfies

- (i) $\emptyset, X \in \mathfrak{M}$.

(ii) If $A \in \mathfrak{M}$, then $X \setminus A \in \mathfrak{M}$.

(iii) If $\{A_n\}_{n=1}^\infty \subset \mathfrak{M}$ is a *mutually disjoint* countable subcollection, then $\bigcup_{n=1}^\infty A_n \in \mathfrak{M}$.

Then prove that $\mathfrak{M}$ is a σ -field.

Theorem 2.2.11 (Carathéodory's Theorem)

Let (X, μ) be a measure space. Then $\mathfrak{M}_\mu$ is a σ -field. Furthermore, if $\{A_n\}_{n=1}^\infty \subset \mathfrak{M}_\mu$ is a *mutually disjoint* countable collection, then we have

$$\mu\left(\bigcup_{n=1}^\infty A_n\right) = \sum_{n=1}^\infty \mu(A_n).$$

proof. Obviously, $\emptyset$ and X belongs to $\mathfrak{M}_\mu$. If $A \in \mathfrak{M}_\mu$, then for every $E \subset X$, we have

$$\mu(E \cap (X \setminus A)) + \mu(E \setminus (X \setminus A)) = \mu(E \setminus A) + \mu(E \cap A) = \mu(E),$$

so $X \setminus A \in \mathfrak{M}_\mu$.

Now, we are going to prove the following: For $A, B \in \mathfrak{M}_\mu$ with $A \cap B = \emptyset$, then $A \cup B \in \mathfrak{M}_\mu$ and for any $E \subset X$,

$$\mu(E \cap (A \cup B)) = \mu(E \cap A) + \mu(E \cap B). \quad (2.1)$$

Since $A \in \mathfrak{M}_\mu$ and both A, B are disjoint, by the definition of measurable sets, we have

$$\begin{aligned} \mu(E \cap (A \cup B)) &= \mu((E \cap A) \cup (E \cap B)) \\ &= \mu([(E \cap A) \cup (E \cap B)] \cap A) + \mu([(E \cap A) \cup (E \cap B)] \setminus A) \\ &= \mu(E \cap A) + \mu(E \cap B). \end{aligned}$$

We will use Exercise 2.2.5 to show $A \cup B \in \mathfrak{M}_\mu$. Fix any $E \subset X$ with $\mu(E) < \infty$. Then since A, B are μ -measurable, and A and B are disjoint,

$$\begin{aligned} \mu(E) &\geq \mu(E \cap A) + \mu(E \setminus A) \\ &\geq \mu(E \cap A) + \mu((E \setminus A) \cap B) + \mu((E \setminus A) \setminus B) \\ &= \mu(E \cap A) + \mu(E \cap (B \setminus A)) + \mu(E \setminus (A \cup B)) \\ &= \mu(E \cap A) + \mu(E \cap B) + \mu(E \setminus (A \cup B)) \\ &\geq \mu(E \cap (A \cup B)) + \mu(E \setminus (A \cup B)). \end{aligned}$$

This proves $A \cup B \in \mathfrak{M}_\mu$.

To show the last condition of the σ -field, we will use Exercise 2.2.10 and prove the last statement at once. Fix $\{A_n\}_{n=1}^\infty$ be a mutually disjoint countable subset of $\mathfrak{M}_\mu$, and $E \subset X$ with $\mu(E) < \infty$. We already proved this case when the collection has only two sets. By the induction over the number of the sets, we find for every $n \geq 2$ that

$$\bigcup_{j=1}^n A_j \in \mathfrak{M}_\mu, \quad \text{and} \quad \sum_{j=1}^n \mu(E \cap A_j) = \mu\left(E \cap \bigcup_{j=1}^n A_j\right) \leq \mu\left(E \cap \bigcup_{j=1}^\infty A_j\right).$$

Therefore,

$$\sum_{j=1}^\infty \mu(E \cap A_j) = \lim_{n \rightarrow \infty} \sum_{j=1}^n \mu(E \cap A_j) \leq \mu\left(E \cap \bigcup_{j=1}^\infty A_j\right) \leq \sum_{j=1}^\infty \mu(E \cap A_j), \quad (2.2)$$

and we find

$$\begin{aligned}
 \mu(E) &\geq \overline{\lim}_{n \rightarrow \infty} \left[\mu \left(E \cap \bigcup_{j=1}^n A_j \right) + \mu \left(E \setminus \bigcup_{j=1}^n A_j \right) \right] \\
 &\geq \overline{\lim}_{n \rightarrow \infty} \left[\mu \left(E \cap \bigcup_{j=1}^n A_j \right) + \mu \left(E \setminus \bigcup_{j=1}^{\infty} A_j \right) \right] \\
 &= \mu \left(E \setminus \bigcup_{j=1}^{\infty} A_j \right) + \overline{\lim}_{n \rightarrow \infty} \mu \left(E \cap \bigcup_{j=1}^n A_j \right) \\
 &= \mu \left(E \setminus \bigcup_{j=1}^{\infty} A_j \right) + \overline{\lim}_{n \rightarrow \infty} \sum_{j=1}^n \mu(E \cap A_j) \\
 &= \mu \left(E \setminus \bigcup_{j=1}^{\infty} A_j \right) + \sum_{j=1}^{\infty} \mu(E \cap A_j) \\
 &= \mu \left(E \setminus \bigcup_{j=1}^{\infty} A_j \right) + \mu \left(E \cap \bigcup_{j=1}^{\infty} A_j \right).
 \end{aligned}$$

This shows that $\bigcup_1^{\infty} A_n \in \mathfrak{M}_{\mu}$, and therefore $\mathfrak{M}_{\mu}$ is a σ -field by Exercise 2.2.10.

Observe that eq. (2.2) can be obtained without the constraint $\mu(E) < \infty$. This yields the last equality by replacing E into X . $\square$

Theorem 2.2.12

Let (X, μ) be a measure space. Then

(i) If A, B are μ -measurable such that $A \subset B$ and $\mu(B) < \infty$, then $\mu(B \setminus A) = \mu(B) - \mu(A)$.

(ii) If A_n are μ -measurable such that

$$A_1 \subset A_2 \subset A_3 \subset \cdots,$$

$$\text{then } \mu\left(\bigcup_1^{\infty} A_n\right) = \lim_{n \rightarrow \infty} \mu(A_n).$$

(iii) If A_n are μ -measurable such that

$$A_1 \supset A_2 \supset A_3 \supset \cdots,$$

$$\text{and } \mu(A_1) < \infty, \text{ then } \mu\left(\bigcap_1^{\infty} A_n\right) = \lim_{n \rightarrow \infty} \mu(A_n).$$

proof. Define $A_0 = \emptyset$ and $B_n = A_n \setminus A_{n-1}$ for $n = 1, 2, 3, \dots$. Then $\bigcup_1^{\infty} B_n = \bigcup_1^{\infty} A_n$ and the collection $\{B_n\}_1^{\infty}$ is mutually disjoint. Thus by applying Theorem 2.2.11,

$$\begin{aligned}
 \mu\left(\bigcup_{n=1}^{\infty} A_n\right) &= \mu\left(\bigcup_{n=1}^{\infty} B_n\right) = \sum_{n=1}^{\infty} \mu(B_n) \\
 &= \lim_{n \rightarrow \infty} \sum_{j=1}^n \mu(B_j) = \lim_{n \rightarrow \infty} \mu\left(\bigcup_{j=1}^n B_j\right) = \lim_{n \rightarrow \infty} \mu(A_n).
 \end{aligned}$$

This proves (ii). To prove (iii), define $B_n = A_1 \setminus A_n$. Then $\bigcup_1^\infty B_n = A_1 \setminus \bigcap_1^\infty A_n$, and

$$B_1 \subset B_2 \subset \cdots.$$

Hence, by (ii), we obtain

$$\mu\left(A_1 \setminus \bigcap_{n=1}^\infty A_n\right) = \mu\left(\bigcup_{n=1}^\infty B_n\right) = \lim_{n \rightarrow \infty} \mu(B_n) = \lim_{n \rightarrow \infty} \mu(A_1 \setminus A_n),$$

and since $\mu(A_1) < \infty$, we can apply (i) and this yields (iii).

(i) is trivial from Theorem 2.2.11 with the simple identity

$$B = A \cup (B \setminus A).$$

This proves the corollary. $\square$

Corollary 2.2.13 (Borel-Cantelli's Lemma)

Let (X, μ) be a measure space and $A_1, A_2, \dots$ be a sequence of μ -measurable subsets of X such that

$$\sum_{n=1}^\infty \mu(A_n) < \infty.$$

Then the set

$$A = \overline{\lim}_{n \rightarrow \infty} A_n = \bigcap_{n=1}^\infty \bigcup_{m=n}^\infty A_m$$

is μ -null.

proof. Since

$$\mu\left(\bigcup_{n=1}^\infty A_n\right) \leq \sum_{n=1}^\infty \mu(A_n) < \infty,$$

we can apply Theorem 2.2.12 (iii) to obtain

$$\mu(A) = \lim_{n \rightarrow \infty} \mu\left(\bigcup_{m=n}^\infty A_m\right) = 0.$$

This proves the corollary. $\square$

We will end this section for more informations about σ -fields. First of all, if $\mathcal{C}$ is a collection of some subsets of X we can generate a σ -field from it.

Definition 2.2.14

Let X be a set and $\mathcal{C}$ be a subset of $\mathcal{P}(X)$. Then the σ -field $\sigma(\mathcal{C})$ generated from $\mathcal{C}$ is defined by

$$\sigma(\mathcal{C}) := \bigcap \{\mathfrak{M} \subset \mathcal{P}(X) : \mathfrak{M} \supset \mathcal{C}, \mathfrak{M} \text{ is a } \sigma\text{-field}\}.$$

That is, $\sigma(\mathcal{C})$ is the smallest σ -field which contains $\mathcal{C}$.

Exercise 2.2.15*

Let X be a nonempty set and $\mathcal{M}$ be a collection of some σ -fields on X . Then the intersection $\bigcap \mathcal{M}$ is also a σ -field. From this fact, $\sigma(\mathcal{C})$ defined in Definition 2.2.14 is really a σ -field.

Exercise 2.2.16*

Let X be a set, $\mathcal{C}$ be some collection of subsets of X and $\mathfrak{M}$ a σ -field such that $\mathcal{C} \subset \mathfrak{M}$. Prove that $\sigma(\mathcal{C}) \subset \mathfrak{M}$.

Exercise 2.2.17

Let X and Y be nonempty sets and $f : X \rightarrow Y$ a function. If $\mathfrak{M}$ is a σ -field on Y then the collection

$$f^{-1}[\mathfrak{M}] := \{f^{-1}[E] : E \in \mathfrak{M}\}$$

is a σ -field on X .

Exercise 2.2.18

Prove that every σ -field is either finite or uncountable.

2.3 Measurable functions

Before defining an integration, we need to say that what kind of functions are “measurable to integrate”. There is a general way to define it, but we first focus for real-valued functions. Everywhere in this section, (X, μ) denotes a measure space. Furthermore, after this moment, we use the fact that $\mathfrak{M}_\mu$ is σ -field (because of Theorem 2.2.11) without any mention.

Definition 2.3.1 (Measurable functions)

Let Y be a topological space. We say that a function $f : X \rightarrow Y$ is μ -measurable if $\{f \in V\}$ is μ -measurable for every open set V in Y . Recall that $\{f \in V\}$ is an abbreviation of $\{x : f(x) \in V\}$.

Below proposition gives an equivalent statement of μ -measurable functions.

Proposition 2.3.2

Let $f : X \rightarrow \mathbb{R}$ be a function, Then the followings are equivalent.

- (i) f is μ -measurable.
- (ii) $\{f \in F\}$ is μ -measurable for every closed subset F of $\mathbb{R}$.
- (iii) $\{f > a\}$ is μ -measurable for every $a \in \mathbb{R}$.
- (iv) $\{f \leq a\}$ is μ -measurable for every $a \in \mathbb{R}$.
- (v) $\{f < a\}$ is μ -measurable for every $a \in \mathbb{R}$.
- (vi) $\{f \geq a\}$ is μ -measurable for every $a \in \mathbb{R}$.
- (vii) $\{f \in I\}$ is μ -measurable for every interval $I \subset \mathbb{R}$.
- (viii) $\{f \in I\}$ is μ -measurable for every open interval $I \subset \mathbb{R}$.
- (ix) $\{f \in I\}$ is μ -measurable for every closed interval $I \subset \mathbb{R}$.

proof. Since $\mathfrak{M}_\mu$ is a σ -field, (iii) and (iv), (v) and (vi) are automatically equivalent, respectively. Now observe that

$$\{f \geq a\} = \bigcap_{n=1}^{\infty} \{f > a - n^{-1}\}, \quad \{f > a\} = \bigcup_{n=1}^{\infty} \{f \geq a + n^{-1}\}.$$

As $\mathfrak{M}_\mu$ is a σ -field, upper relation yields that (iii) and (vi) are equivalent. Hence, (iii)-(vi) are all equivalent.

Notice that clearly all (iii)-(vi) imply (vii), and (vii) implies (viii) and (ix). In addition, since

$$\{f > a\} = \bigcup_{n=1}^{\infty} \{a < f < n\}, \quad \{f \geq a\} = \bigcup_{n=1}^{\infty} \{a \leq f \leq n\},$$

(viii) implies (iii), and (ix) implies (vi).

Every open set in $\mathbb{R}$ can be expressed by a countable union of open intervals in $\mathbb{R}$. Thus (viii) implies (i). Also, clearly (i) implies (viii). Finally, observe that clearly (i) and (ii) are equivalent.

This finishes the proof. $\square$

Now we are going to study μ -measurable $[-\infty, \infty]$ -valued functions. Such functions is needed if one consider supremum or infimum of functions. Recall that the topology on $[-\infty, \infty]$ is given by an order topology where $-\infty \leq a$ and $\infty \geq a$ for every $a \in [-\infty, \infty]$. Equivalently, following metric $d(x, y) = |\arctan x - \arctan y|$ which makes the space complete and generates the same topology described in above. Here, we define $\arctan \infty := \pi/2$, $\arctan(-\infty) := -\pi/2$.

Proposition 2.3.3

Let $f : X \rightarrow [-\infty, \infty]$ is μ -measurable if and only if for every open subset V of $\mathbb{R}$, $\{f \in V\}$, $\{f = \infty\}$, and $\{f = -\infty\}$ are μ -measurable.

proof. First, assume that f is μ -measurable, and fix any open subset V of $\mathbb{R}$. Since both $[-\infty, a)$ and $(a, \infty]$ are open in $[-\infty, \infty]$ for every $a \in \mathbb{R}$, we have

$$\{f = \infty\} = \bigcap_{n=1}^{\infty} \{f \in (n, \infty]\} \in \mathfrak{M}_\mu, \quad \{f = -\infty\} = \bigcap_{n=1}^{\infty} \{f \in [-\infty, -n)\} \in \mathfrak{M}_\mu.$$

Notice that V is also open in $[-\infty, \infty]$ because $\mathbb{R}$ is a subspace of $[-\infty, \infty]$. Hence, $\{f \in V\}$ is clearly μ -measurable.

Now assume the converse, and W be any open subset of $[-\infty, \infty]$. The only cases we need to consider is that either $-\infty \in W$ or $\infty \in W$, without loss of generality, assume that $\infty \in W$ and $-\infty \notin W$. Since $W \cap \mathbb{R}$ is open in $\mathbb{R}$, by our assumption, $\{f \in W \cap \mathbb{R}\}$ is μ -measurable. Therefore,

$$\{f \in W\} = \{f \in W \cap \mathbb{R}\} \cup \{f = \infty\} \in \mathfrak{M}_\mu.$$

This proves that f is μ -measurable. $\square$

We already know that if f and g are real-valued continuous function on the real line, then both $f + g$ and fg are also continuous. Then one can question about for measurability. Similar for continuity, if f and g are real-valued μ -measurable functions, then both $f + g$ and fg are μ -measurable.

Proposition 2.3.4

Let $f, g : X \rightarrow \mathbb{R}$ be μ -measurable functions. Then $-f$, $f + g$ and fg are μ -measurable.

proof. The measurability of $-f$ is immediately follows from Proposition 2.3.2. Fix $a \in \mathbb{R}$. Then observe that

$$\{f + g > a\} = \{f > a - g\} = \bigcup_{q \in \mathbb{Q}} \{f > q\} \cap \{g > a - q\}.$$

Since f and g are μ -measurable, and $\mathfrak{M}_\mu$ is a σ -field, that identity implies that $\{f + g > a\}$ is μ

measurable. As $a \in \mathbb{R}$ is arbitrary, we can conclude that $f + g$ is μ -measurable. To prove that fg is μ -measurable, we use the following formula

$$fg = \frac{1}{4}(f + g)^2 - \frac{1}{4}(f - g)^2.$$

Hence, it suffices to show that f^2 is μ -measurable. Fix $a \in \mathbb{R}$. If $a < 0$ then clearly $\{f^2 > a\} = X$, so we can further assume that $a \geq 0$. In this case,

$$\{f^2 > a\} = \{|f| > \sqrt{a}\} = \{f \geq 0, f > \sqrt{a}\} \cup \{f < 0, f < -\sqrt{a}\},$$

and the right hand side set is indeed μ -measurable by Proposition 2.3.2. $\square$

Exercise 2.3.5*

Let $f : X \rightarrow \mathbb{R}$ be a function, and we define the positive part and the negative part of f by

$$f_+ := f \vee 0, \quad f_- := (-f) \vee 0.$$

Prove that if f is μ -measurable, so are f_+ and f_- .

In addition, if $g : X \rightarrow \mathbb{R}$ is another μ -measurable function, prove that both $f \vee g$ and $f \wedge g$ are μ -measurable.

In the proof of Proposition 2.3.4, we actually proved that f^2 and $|f|$ are μ -measurable provided that f is μ -measurable. One can notice that from Proposition 2.3.2, we can prove the following result.

Proposition 2.3.6

Let Y and Z are topological spaces, $f : X \rightarrow Y$ be μ -measurable function, and $\phi : Y \rightarrow Z$ be a continuous function. Then $\phi \circ f$ is μ -measurable.

proof. Fix any open set U in Z . Being continuous, $\phi^{-1}[U]$ is also open in Y . Therefore, since

$$\{\phi \circ f \in U\} = \{f \in \phi^{-1}[U]\},$$

we can conclude that $\phi \circ f$ is μ -measurable. $\square$

If f is μ -measurable, we can obtain very interesting result which used in the future.

Proposition 2.3.7

Let $f : X \rightarrow \mathbb{R}$ be μ -measurable function. Then the function $g(t) = \mu\{f > t\}$ is decreasing right continuous on $(0, \infty)$. Furthermore, g is LSC on $(0, \infty)$.

proof. For $0 < s < t < \infty$, since $\{f > t\} \subset \{f > s\}$, we have $\mu\{f > t\} \leq \mu\{f > s\}$. Thus the function g is decreasing. To prove the right continuity, fix $t_0 \in (0, \infty)$ and sequence $(t_n)_n$ on $(0, \infty)$ such that $t_n \downarrow t_0$. Since f is μ -measurable, Theorem 2.2.12 and the identity

$$\bigcup_{n=1}^{\infty} \{f > t_n\} = \{f > t\}$$

implies $g(t_n) \rightarrow g(t_0)$ as $n \rightarrow \infty$. This proves the right continuity.

To prove that g is LSC, fix $t_0 \in (0, \infty)$ and real y such that $y < g(t_0)$. Since g is right continuous, there exists $\delta > 0$ such that $g(t) > y$ holds for every $t \in [t_0, t_0 + \delta)$. Since g is decreasing, actually $g(t) > y$ holds for every $|t - t_0| < \delta$. Hence we proved that g is LSC at t_0 since y is arbitrary. Now recall that we take any point t_0 from $(0, \infty)$, thus the proposition is proved. $\square$

On the measure theory, although a proposition $\mathbf{P}(x)$ does not satisfies for *all* $x \in X$, it might be a case that the collection of all points $x \in X$ at which the proposition $\mathbf{P}(x)$ is false has measure zero. In other words,

$$\mu\{x \in X : \mathbf{P}(x) \text{ is false}\} = 0.$$

Definition 2.3.8 (Almost Everywhere)

Let (X, μ) be a measure space and $\mathbf{P}(x)$ is a proposition with a parameter $x \in X$. We say that $\mathbf{P}(x)$ is true *almost everywhere* if

$$\mu\{x \in X : \mathbf{P}(x) \text{ is false}\} = 0.$$

We also use an abbreviation: $\mathbf{P}(x)$ is true (μ -a.e.), or just $\mathbf{P}(x)$ (a.e.) if there is no confusion of the measure.

Let us consider two functions f and g on X . If we know that f is μ -measurable, and $f = g$ (a.e.), what can we say about g then? Furthermore, if $(f_n)_n$ is a sequence of μ -measurable functions and it converges to f (a.e.). Then what can we sat about f then? Below two proposition gives the answer. Especially, Proposition 2.3.10 is originated by [2].

Proposition 2.3.9

Let Y be a topological space, and f and g are functions from X into Y such that $f = g$ (a.e.). If f is μ -measurable, so is g .

proof. Fix any open set V in Y . Then we have

$$\{g \in V\} = (\{f = g\} \cap \{f \in V\}) \cup \{f \neq g\}.$$

Since $f = g$ (a.e.), $\mu\{f \neq g\} = 0$, thus the set $\{f \neq g\}$ is μ -measurable by Exercise 2.2.6. Using Theorem 2.2.11, the set $\{f = g\}$ is also μ -measurable. Since we assumed that f is μ -measurable, the set $\{f \in V\}$ is also μ -measurable because of the definition of the measurable functions. Combining these, we therefore conclude that $\{g \in V\}$ is μ -measurable. Since V can be any open sets in Y , we proved that g is μ -measurable. $\square$

Proposition 2.3.10

Let Y be a perfectly normal space, and let $(f_n)_n$ be a sequence of μ -measurable functions from X into Y . If $f_n \rightarrow f$ (a.e.) for some $f : X \rightarrow Y$, then f is also μ -measurable.

proof. Let N be the collection of all $x \in X$ such that $f_n(x)$ converges to $f(x)$. Then $\mu(X \setminus N) = 0$ because $f_n \rightarrow f$ almost everywhere.

Fix any open set V in Y . By Theorem 1.4.10, there exists a continuous function $\phi : X \rightarrow [0, 1]$ such that $V = \{\phi > 0\}$. Define $F_n = \{\phi \geq n^{-1}\}$. Clearly F_n are closed because ϕ is continuous, and $\bigcup_n F_n = V$.

Fix any $x \in \{f \in V\} \cap N$. Take a positive integer k such that $f(x) \in \{\phi > k^{-1}\}$. Since $f_n(x) \rightarrow f(x)$ as $n \rightarrow \infty$, we can take sufficiently large m so that $f_n(x) \in \{\phi > k^{-1}\}$ for every $n \geq m$. From this fact, we obtain

$$\begin{aligned} \{f \in V\} \cap N &= N \cap \bigcup_{k=1}^{\infty} \{\phi \circ f > k^{-1}\} \\ &\subset N \cap \bigcup_{k=1}^{\infty} \bigcup_{m=1}^{\infty} \bigcap_{n=m}^{\infty} \{\phi \circ f_n > k^{-1}\} \end{aligned}$$

$$\begin{aligned} &\subset N \cap \bigcup_{k=1}^{\infty} \bigcup_{m=1}^{\infty} \bigcap_{n=m}^{\infty} \{f_n \in F_k\} \\ &=: N \cap E. \end{aligned}$$

On the other hand, if $x \in N \cap E$, that means, there exist k and m such that $f_n(x) \in F_k$ for every $n \geq m$ and $f_n(x) \rightarrow f(x)$ as $n \rightarrow \infty$. Since F_k is closed, this implies $f(x) \in F_k$. Hence, we conclude that

$$\{f \in V\} \cap N = N \cap \bigcup_{k=1}^{\infty} \bigcup_{m=1}^{\infty} \bigcap_{n=m}^{\infty} \{f_n \in F_k\}. \quad (2.3)$$

As $X \setminus N$ is μ -null set, it is μ -measurable, so N is also μ -measurable. In addition, each f_n are μ -measurable, and F_k are closed in Y . This implies that $\{f_n \in F_k\}$ is μ -measurable for each n and k . Therefore, eq. (2.3) gives that $\{f \in V\} \cap N$ is μ -measurable.

Remark 2.3.11

If one consider a measure space $(X, \mathfrak{M}, \mu)$ described in Remark 2.2.7, There is no guarantee that although $f = g$ (a.e.), f is measurable implies g is also measurable. This is because, unlike our setting, “measurability” is given from σ -field $\mathfrak{M}$ not from a measure μ . Which means, not only we cannot know the measurability of $\{f \neq g\}$, the formula $\mu\{f \neq g\} = 0$ is actually not defined because in general, the domain of the measure μ is $\mathfrak{M}$.

However, we started from a measure defined at Definition 2.2.1 and define “measurability” using the measure. Since the domain of our measure is the power set of the space, $\mu\{f \neq g\}$ makes sense in the first place. Hence Proposition 2.3.9 makes sense also for our setting. This is the one of the benefit to define a measure like Definition 2.2.1 not like in Remark 2.2.7.

Remark 2.3.12

Proposition 2.3.10 does not holds even when Y is completely normal. Consider $Y = [0, 1] \times [0, 1]$ with lexicographic order. One can easily check that Y is then a completely normal space (see [7]).

We have been studied the definition of μ -measurable functions. We end this section by showing one of the most important theorem about μ -measurable functions. Before that we need a definition.

Definition 2.3.13 (Simple functions)

We say that a function is called *simple* if it takes only finitely many values.

2.4 Integrations

Now we are ready to define a Lebesgue integration. Everywhere in this section, (X, μ) denotes a measure space.

Suppose that a function $f : X \rightarrow [0, \infty)$ is given and we want to measure the area of the region $\Gamma = \{(x, y) : 0 < y \leq f(x)\}$. Fix a positive integer n and “slice” Γ by sets of type

$$\Gamma_{n,k} = \{x : f(x) > k2^{-n}\} \times (k2^{-n}, (k+1)2^{-n}], \quad k = 0, 1, \dots, n2^n - 1.$$

Since the shape of $\Gamma_{n,k}$ is a rectangle, intuitively we can “define” the area of $\Gamma_{n,k}$ by $2^{-n}\mu\{x : f(x) > k2^{-n}\}$. Thus it is reasonable to “define” the area of Γ by

$$\lim_{n \rightarrow \infty} \sum_{k=1}^{n2^n} \frac{1}{2^n} \mu\{x : f(x) > k2^{-n}\}$$

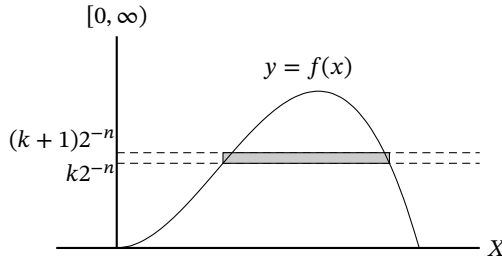


Figure 2.1: Definition of the Lebesgue integration

if the limit exists. For this reason, we can define an integration by following.

Definition 2.4.1 (Lebesgue Integration)

Let $f : X \rightarrow \mathbb{R}$. Suppose that at least one of the expressions

$$\int_0^\infty \mu\{f_+ > t\}dt, \quad \int_0^\infty \mu\{f_- > t\}dt \quad (2.4)$$

is finite (see Exercise 2.3.5 for their definitions). Then we define

$$\int_X f(x)\mu(dx) := \int_0^\infty \mu\{f_+ > t\}dt - \int_0^\infty \mu\{f_- > t\}dt.$$

If f is complex valued and both integrals

$$\int_X \operatorname{Re} f(x)\mu(dx), \quad \int_X \operatorname{Im} f(x)\mu(dx)$$

are meaningful, then we define

$$\int_X f(x)\mu(dx) = \int_X \operatorname{Re} f(x)\mu(dx) + i \int_X \operatorname{Im} f(x)\mu(dx)$$

If $E \subset X$ is a set, then we define the Lebesgue integral on E by

$$\int_E f(x)\mu(dx) := \int_X \mathbb{1}_E(x)f(x)\mu(dx).$$

Sometimes we also write $\int_X f(x)\mu(dx)$ by $\int_X f(x)d\mu(x)$, or $\int_X f d\mu$. In addition, we say that a function f is μ -integrable if $\int_X |f|d\mu < \infty$.

Remark 2.4.2

In the definition of the Lebesgue integral, we need to justify that eq. (2.4) is meaningful. If f is nonnegative, we must justify that the function $t \mapsto \mu\{f > t\}$ is Riemann integrable. If $\mu\{f > t\} = \infty$ for some $t \in (0, \infty)$, then just define

$$\int_0^\infty \mu\{f > t\}dt = \infty.$$

Now let us assume that $\mu\{f > t\} < \infty$ for every $t \in (0, \infty)$. By remembering Proposition 2.3.7, the function $t \mapsto \mu\{f > t\}$ is decreasing. This implies that it is bounded on every closed interval $[\epsilon, T] \subset (0, \infty)$, hence it is Riemann integrable. Hence, we understood the integral on $(0, \infty)$ as

an improper Riemann integral:

$$\int_0^\infty \mu\{f > t\}dt = \lim_{\substack{T \uparrow \infty \\ \epsilon \downarrow 0}} \int_\epsilon^T \mu\{f > t\}dt.$$

This shows that eq. (2.4) is well-defined.

We first study the basic properties of the Lebesgue integral. Below theorem holds for *every* functions, that is, a function is not need to be μ -measurable.

Theorem 2.4.3

Let f and g are functions from X into $\mathbb{R}$. Then

(i) If $f \geq 0$ (a.e.), then

$$\int_X f d\mu = \int_0^\infty \mu\{f > t\}dt.$$

(ii) If $f \leq g$ (a.e.), and f_- is μ -integrable, then

$$\int_X f d\mu \leq \int_X g d\mu.$$

Moreover, if g_+ is μ -integrable, then both f and g are μ -integrable.

(iii) If $\int_X f d\mu$ is meaningful, then

$$\left| \int_X f d\mu \right| \leq \int_X |f| d\mu.$$

(iv) For every $\alpha > 0$,

$$\mu\{|f| > \alpha\} \leq \frac{1}{\alpha} \int_X |f| d\mu.$$

proof. (i) Since $f \geq 0$ almost everywhere, we have $\mu\{f < 0\} = 0$. Notice that on $\{f < 0\}$, $f_- > 0$. This implies that $\{f_- > t\} \subset \{f < 0\}$ for every $t \in (0, \infty)$, and thus

$$0 \leq \int_0^\infty \mu\{f_- > t\}dt = \lim_{\substack{T \uparrow \infty \\ \epsilon \downarrow 0}} \int_\epsilon^T \mu\{f_- > t\}dt \leq \lim_{\substack{T \uparrow \infty \\ \epsilon \downarrow 0}} \int_\epsilon^T \mu\{f < 0\}dt = 0.$$

On the other hand, by the definition of f_+ , we have $\{f_+ > t\} = \{f > t\}$ for every $t \in (0, \infty)$. Therefore,

$$\int_X f d\mu = \int_0^\infty \mu\{f_+ > t\}dt - \int_0^\infty \mu\{f_- > t\}dt = \int_0^\infty \mu\{f > t\}dt.$$

(ii) First of all, assume that both f and g are nonnegative. Observe that for every $t \geq 0$ that

$$\begin{aligned} \{f > t\} &= (\{f > g\} \cap \{f > t\}) \cup (\{f \leq g\} \cap \{f > t\}) \\ &\subset \{f > g\} \cup (\{f \leq g\} \cap \{g > t\}) \\ &\subset \{f > g\} \cup \{g > t\}. \end{aligned}$$

Since $f \leq g$ (a.e.), $\mu\{f > g\} = 0$. Hence, we have

$$\mu\{f > t\} \leq \mu\{f > g\} + \mu\{g > t\} = \mu\{g > t\}.$$

By integrating both side over t therefore proves that (also use the result of the previous one)

$$\int_X f d\mu = \int_0^\infty \mu\{f > t\} dt \leq \int_0^\infty \mu\{g > t\} dt = \int_X g d\mu.$$

In general, define $N = \{f \leq g\}$. Then notice that $\mu(X \setminus N) = 0$. Observe that the function $x \mapsto x \vee 0$ is an increasing function. Thus on N , $f \leq g$ implies $f_+ \leq g_+$. Similarly, since $-f \geq -g$ on N , this implies that $f_- \geq g_-$. Since f_+ , f_- , g_+ , and g_- are all nonnegative functions, by the above result, we have

$$\int_X f_+ d\mu \leq \int_X g_+ d\mu, \quad \int_X g_- d\mu \leq \int_X f_- d\mu. \quad (2.5)$$

Adding both inequalities, therefore, gives

$$\int_X f_+ d\mu + \int_X g_- d\mu \leq \int_X f_- d\mu + \int_X g_+ d\mu.$$

Since f_- is μ -integrable, g_- is also μ -integrable by eq. (2.5). Then by obvious subtraction in both side gives $\int_X f d\mu \leq \int_X g d\mu$. Furthermore, if g_+ is μ -integrable, then f_+ is also μ -integrable by eq. (2.5). This proves that both f and g are μ -integrable.

(iii) Since $f_+ \leq |f|$ and $f_- \leq |f|$, the previous result gives

$$\int_X f_+ d\mu \leq \int_X |f| d\mu, \quad \int_X f_- d\mu \leq \int_X |f| d\mu.$$

Recall the obvious inequality: if $a, b \geq 0$, then $|a - b| \leq a \vee b$. This gives that

$$\left| \int_X f d\mu \right| = \left| \int_X f_+ d\mu - \int_X f_- d\mu \right| \leq \left(\int_X f_+ d\mu \right) \vee \left(\int_X f_- d\mu \right) \leq \int_X |f| d\mu.$$

(iv) It suffices to assume that f is nonnegative everywhere. Then since $t \mapsto \mu\{f > t\}$ is non-negative decreasing function, we have together with (i),

$$\int_X f d\mu = \int_0^\infty \mu\{f > t\} dt \geq \int_0^\alpha \mu\{f > t\} dt \geq \alpha \mu\{f > \alpha\}.$$

The theorem is now proved. $\square$

Remark 2.4.4

If f is complex-valued, then Theorem 2.4.3 (iii) is false in general. Consider $X = \{a, b\}$, and define a measure μ by

$$\mu(\emptyset) = 0, \quad \mu\{a\} = \mu\{b\} = \mu(X) = 1.$$

Now consider $f : X \rightarrow \mathbb{C}$ by $f(a) = 1$, and $f(b) = i$. Then first of all, we have

$$\int_X \operatorname{Re} f d\mu = \int_0^\infty \mu\{\operatorname{Re} f > t\} dt = \int_0^1 \mu\{a\} dt = 1,$$

$$\int_X \operatorname{Im} f d\mu = \int_0^\infty \mu\{\operatorname{Im} f > t\} dt = \int_0^1 \mu\{b\} dt = 1.$$

Thus we have $\int_X f d\mu = 1 + i$. On the other hand, since $|f| = 1$,

$$\int_X |f| d\mu = \int_0^\infty \mu\{1 > t\} dt = \int_0^1 \mu(X) dt = 1.$$

Since $|1 + i| = \sqrt{2}$, we find

$$\left| \int_X f d\mu \right| = \sqrt{2} > 1 = \int_X |f| d\mu,$$

thus Theorem 2.4.3 (iii) does not hold for complex-valued functions, in general. Notice that actually, f in the above example is not μ -measurable.

Now we are going to see three important theorems in real analysis. They give the condition when an integral and a limit can be interchanged. First of all, we will study the monotone convergence theorem. To prove this theorem, we need one lemma whose proof is in the appendix.

Lemma 2.4.5

Let f be a nonnegative decreasing function on an interval $[a, b]$, where $-\infty < a < b < \infty$. Prove that for every $\epsilon > 0$, there exists a continuous function g on $[a, b]$ such that $0 \leq g \leq f$, and

$$\int_a^b f(x) dx < \epsilon + \int_a^b g(x) dx,$$

where the integral is understood as the *Riemann integral* sense.

Theorem 2.4.6 (The Monotone Convergence Theorem)

Let f_n be a sequence of nonnegative μ -measurable functions such that $f_n \uparrow f$ as $n \rightarrow \infty$. That is, $0 \leq f_1 \leq f_2 \leq \dots \leq f$, and f_n converges to f as $n \rightarrow \infty$. Then

$$\lim_{n \rightarrow \infty} \int_X f_n d\mu = \int_X f d\mu.$$

proof. By Theorem 2.4.3 (ii), as $f_n \leq f_{n+1} \leq f$ on X for every n , we have

$$\int_X f_n d\mu \leq \int_X f_{n+1} d\mu \leq \int_X f d\mu.$$

Hence the sequence $\int_X f_n d\mu$ increases, and we obtain

$$\lim_{n \rightarrow \infty} \int_X f_n d\mu \leq \int_X f d\mu.$$

To prove the converse, it suffices to assume that (recall that the sequence is increasing)

$$\sup_n \int_X f_n d\mu = \lim_{n \rightarrow \infty} \int_X f_n d\mu < \infty.$$

By Theorem 2.4.3 (iv), upper assumption implies that for every $t \in (0, \infty)$,

$$\sup_n \mu\{f_n > t\} < \infty. \quad (2.6)$$

On the other hand, notice that $\{f_n > t\} \subset \{f_{n+1} > t\}$ and $\{f > t\} = \bigcup_{n=1}^\infty \{f_n > t\}$ for every n and $t \in (0, \infty)$ because $f_n \uparrow f$ as $n \rightarrow \infty$. In addition, since f_n is μ -measurable, f is also μ -measurable

by Proposition 2.3.10. Thus using Theorem 2.2.12 (ii), $\mu\{f_n > t\} \rightarrow \mu\{f > t\}$ as $n \rightarrow \infty$ for every $t \in (0, \infty)$. Because of eq. (2.6), $\mu\{f > t\} < \infty$ for every $t \in (0, \infty)$. Since $t \mapsto \mu\{f > t\}$ is decreasing, the function is Riemann integrable on every compact subinterval of $(0, \infty)$.

Fix $\epsilon, \eta > 0$ and $T \in (0, \infty)$, and take a nonnegative continuous function g on $[\epsilon, T]$ such that $g(t) \leq \mu\{f > t\}$ for every $t \in [\epsilon, T]$, and

$$\int_{\epsilon}^T \mu\{f > t\} dt < \eta + \int_{\epsilon}^T g(t) dt$$

by Lemma 2.4.5.

Now consider $g_n(t) = g(t) \wedge \mu\{f_n > t\}$. Then g_n is LSC for each n because of Proposition 2.3.7 and Proposition 1.3.6. In addition $g_n \uparrow g$ as $n \rightarrow \infty$ on $[\epsilon, T]$. Since g is continuous and $[\epsilon, T]$ is compact, by Theorem 1.6.9, $g_n \rightarrow g$ uniformly on $[\epsilon, T]$. This together with Theorem 2.4.3 implies

$$\begin{aligned} \int_{\epsilon}^T \mu\{f > t\} dt &< \eta + \int_{\epsilon}^T g(t) dt \\ &= \eta + \lim_{n \rightarrow \infty} \int_{\epsilon}^T g_n(t) dt \\ &\leq \eta + \lim_{n \rightarrow \infty} \int_{\epsilon}^T \mu\{f_n > t\} dt \\ &\leq \eta + \lim_{n \rightarrow \infty} \int_0^{\infty} \mu\{f_n > t\} dt \\ &= \eta + \lim_{n \rightarrow \infty} \int_X f_n d\mu. \end{aligned}$$

Letting $T \uparrow \infty$, $\epsilon \downarrow 0$, and $\eta \downarrow 0$ in order, we proved the converse inequality. Therefore the proof is finished. $\square$

2.5 Borel sets and Borel measures

If X is a topological space, the topology τ of X is one of subsets of $\mathcal{P}(X)$. This means we can consider the σ -field $\sigma(\tau)$. Such σ -field has a special name and used in several places in measure theory.

Definition 2.5.1 (Borel σ -fields, Borel measure)

Let X be a topological space and τ be its topology. Then the σ -field $\sigma(\tau)$ is called the *Borel σ -field* and denoted by $\mathfrak{B}(X)$.

Every set belongs to the Borel σ -field is called by a *Borel set*.

If X has a measure μ such that all Borel sets are μ -measurable, then we say that μ is a *Borel measure*, or μ is *Borel* in short.

Lemma 2.5.2

Let X be a topological space and μ is a measure on X . The μ is Borel if and only if every open set is μ -measurable if and only if every closed set is μ -measurable.

proof. By Theorem 2.2.11, $\mathfrak{M}_{\mu}$ is a σ -field. Hence, every open set is μ -measurable means that every open set belongs to $\mathfrak{M}_{\mu}$, so $\mathfrak{B}(X) \subset \mathfrak{M}_{\mu}$ because of Exercise 2.2.16. In addition, the fact that $\mathfrak{M}_{\mu}$ is a σ -field clearly implies that every open set is μ -measurable if and only if every closed set is

μ -measurable. Finally, if μ is Borel measurable, then $\mathfrak{B}(X)$ is contained in $\mathfrak{M}_\mu$. By the definition of the Borel σ -field, clearly every open set is then μ -measurable. $\square$

Checking whether a measure is Borel is difficult. However if the space has a metric, then there is an easy way to verify it. The statement of the theorem and its proof can be found also in [5].

Theorem 2.5.3

Let X be a metric space with a metric d , and μ be a measure on X . For any $x \in X$ and subsets A and B of X , define

$$d(x, B) := \inf\{d(x, y) : y \in B\}, \quad d(A, B) := \inf\{d(x, B) : x \in A\} \quad (\inf \emptyset := \infty).$$

If $\mu(A \cup B) = \mu(A) + \mu(B)$ whenever $d(A, B) > 0$, then μ is a Borel measure.

proof. By Lemma 2.5.2 it suffices to show that every closed set is μ -measurable. Fix any closed set C in X and any subset $A \subset X$. By Exercise 2.2.5, we can further assume that $\mu(A) < \infty$. For each $n = 1, 2, 3, \dots$, consider

$$C_n := \{x \in X : d(x, C) \leq 2^{-n}\}.$$

One can easily show that C_n is closed because the function $d(\cdot, C)$ is continuous, and $\bigcap_{n=1}^{\infty} C_n = C$ because C is closed. First of all, observe that $d(A \setminus C_n, A \cap C) > 0$ for all n . Indeed, for any $x \in A \setminus C_n$, we have

$$d(x, A \cap C) \geq d(x, C) > \frac{1}{2^n}.$$

Thus, taking an infimum over x , we obtain $d(A \setminus C_n, A \cap C) \geq 2^{-n} > 0$. Then our assumption about μ gives

$$\mu(A) \geq \mu((A \setminus C_n) \cup (A \cap C)) = \mu(A \setminus C_n) + \mu(A \cap C). \quad (2.7)$$

Claim that $\mu(A \setminus C_n) \rightarrow \mu(A \setminus C)$ as $n \rightarrow \infty$. Beware that we cannot use Theorem 2.2.12 because we do not know their measurability. Define E_n for $n = 1, 2, \dots$ by

$$E_n := A \cap (C_n \setminus C_{n+1}) = \{x \in A : 2^{-n-1} < d(x, C) \leq 2^{-n}\}.$$

First of all, observe that $d(E_n, E_{n+2}) > 0$ for each $n = 1, 2, \dots$. Indeed, for any $x \in E_n$, $y \in E_{n+2}$, and $c \in C$,

$$\frac{1}{2^{n+1}} < d(x, c) \leq d(x, y) + d(y, c).$$

Taking an infimum over extreme parts of the inequality, we obtain

$$\frac{1}{2^{n+1}} \leq d(x, y) + d(y, C) \leq d(x, y) + \frac{1}{2^{n+2}},$$

and thus $d(x, y) \geq 2^{-n-2}$. Taking infimums over x and y therefore gives $d(E_n, E_{n+2}) \geq 2^{-n-2} > 0$. Such fact and our assumption gives

$$\sum_{n=1}^N \mu(E_{2n}) + \mu(E_{2n-1}) = \mu\left(\bigcup_{n=1}^N E_{2n}\right) + \mu\left(\bigcup_{n=1}^N E_{2n-1}\right) \leq 2\mu(A).$$

Letting $N \rightarrow \infty$ gives $\sum_{n=1}^{\infty} \mu(E_n) \leq 2\mu(A)$. Recall that we assumed that $\mu(A)$ is finite. From this fact and since $C = \bigcap_{n=1}^{\infty} C_n$, we can now show our claim by

$$\overline{\lim}_{n \rightarrow \infty} \mu(A \setminus C_n) \leq \mu(A \setminus C)$$

$$\begin{aligned}
&\leq \lim_{n \rightarrow \infty} \left[\mu(A \setminus C_n) + \sum_{m=n}^{\infty} \mu(E_m) \right] \\
&= \lim_{n \rightarrow \infty} \mu(A \setminus C_n) + \lim_{n \rightarrow \infty} \sum_{m=n}^{\infty} \mu(E_m) \\
&= \lim_{n \rightarrow \infty} \mu(A \setminus C_n).
\end{aligned}$$

Therefore, letting $n \rightarrow \infty$ on eq. (2.7) gives

$$\mu(A) \geq \mu(A \setminus C) + \mu(A \cap C),$$

hence C is μ -measurable because of Exercise 2.2.5. This finishes the proof. $\square$

From Theorem 2.5.3 we can construct a canonical Borel measure on any metric spaces.

Theorem 2.5.4

Let (X, d) be a metric space and $f : (0, \infty) \rightarrow [0, \infty)$ be a (monotonically) increasing function such that $f(0+) = 0$. Then the function $\ell_{X,d,f} : \mathcal{P}(X) \rightarrow [0, \infty]$ defined by

$$\ell_{X,d,f}(E) := \inf \left\{ \sum_{n=1}^{\infty} f(r_n) : E \subset \bigcup_{n=1}^{\infty} B(a_n, r_n), a_n \in X, r_n > 0 \right\} \quad (\inf \emptyset := \infty)$$

is a Borel measure on X .

proof. By Exercise 2.5.5, $\ell_{X,d,f}$ is a measure.

To show that $\ell_{X,d,f}$ is a Borel measure, we are going to use Theorem 2.5.3. Fix two subsets A and B such that $d(A, B) > 0$. If $\ell_{X,d,f}(A \cup B) = \infty$, there is nothing to prove, so assume that $\ell_{X,d,f}(A \cup B) < \infty$. Put $\delta = d(A, B)/4$ and take a sequence $(B(a_n, r_n))_1^\infty$ of open balls such that $A \cup B \subset \bigcup_{n=1}^{\infty} B(a_n, r_n)$. Define

$$\mathcal{A} := \{n : A \cap B(a_n, r_n) \neq \emptyset\}, \quad \mathcal{B} := \{n : B \cap B(a_n, r_n) \neq \emptyset\}.$$

Observe that $\mathcal{A}$ and $\mathcal{B}$ are disjoint. Indeed, if there exists $N \in \mathcal{A} \cap \mathcal{B}$, we can take $a \in A \cap B(a_N, r_N)$ and $b \in B \cap B(a_N, r_N)$. Then we have

$$4\delta = d(A, B) \leq d(a, b) \leq 2r_N \leq 2\delta,$$

which makes a contradiction.

Now we have

$$A \subset \bigcup_{n \in \mathcal{A}} B(a_n, r_n), \quad B \subset \bigcup_{n \in \mathcal{B}} B(a_n, r_n),$$

and

$$\sum_{n=1}^{\infty} f(r_n) = \sum_{n \in \mathcal{A}} f(r_n) + \sum_{n \in \mathcal{B}} f(r_n) \geq \ell_{X,d,f}(A) + \ell_{X,d,f}(B).$$

Since f is increasing, taking an infimum over sequence of such open balls, we obtain $\ell_{X,d,f}(A \cup B) \geq \ell_{X,d,f}(A) + \ell_{X,d,f}(B)$. Therefore $\ell_{X,d,f}$ is a Borel measure because of Theorem 2.5.3. $\square$

Exercise 2.5.5*

Prove that $\ell_{X,d,f}$ defined in Theorem 2.5.4 is a measure on X .

Definition 2.5.6

Let (X, d) be a metric space. We call $\ell_{X,d,f}$ defined in Theorem 2.5.4 by the *canonical Borel measure on X generated by f* .

Now we are going to look the specific measure on $\mathbb{R}^d$ in daily use, the Lebesgue measure. If ℓ denotes the Lebesgue measure on $\mathbb{R}^d$ whose definition is presented in later, then we obtain $\mathfrak{B}(\mathbb{R}^d) \subset \mathfrak{M}_\ell$ because of Theorem 2.5.3. However both σ -fields are not equal. Furthermore, we also going to prove that $\mathfrak{M}_\ell \neq \mathcal{P}(\mathbb{R}^d)$ at the end of this section. First of all, we should learn what is the Lebesgue measure on $\mathbb{R}^d$.

Definition 2.5.7 (Lebesgue measure on $\mathbb{R}^d$)

Let $d \geq 1$ be an integer. The *Lebesgue measure* on $\mathbb{R}^d$ is denoted by ℓ defined by $\ell := \ell_{\mathbb{R}^d, \delta, f}$ where $f(x) = (2x)^d$ and

$$\delta(x, y) = \max_{1 \leq j \leq d} |x_j - y_j|.$$

The idea of defining the Lebesgue measure in Definition 2.5.7 is that we want the area of the cube

$$[a_1, b_1] \times \cdots \times [a_d, b_d]$$

by $(b_1 - a_1)(b_2 - a_2) \cdots (b_d - a_d)$, which is what we expect in nature. However this is not trivial directly from the definition. Below proposition proves this fact.

Proposition 2.5.8

$$\ell([a_1, b_1] \times \cdots \times [a_d, b_d]) = (b_1 - a_1)(b_2 - a_2) \cdots (b_d - a_d).$$

proof. **[Fill Here!!!]**

2.6 Hints for Exercises

[Exercise 2.3.5] $f \wedge 0 = -((-f) \vee 0)$ and $f \vee 0 = (|f| + f)/2$

Filters

A.1 Filters and Ultrafilters

We already studied one of the generalization of sequences: nets. In here, we are going to study about filters. One can see the filter theory at [1, 2, 7].

First, we need to see what are filters. Note that filters are defined even if X has no topology.

Definition A.1.1 (Filters)

Let X be a nonempty set. A *filter* $\mathcal{F}$ is a nonempty collection of some subsets of X satisfies the following conditions.

- (i) $\emptyset \notin \mathcal{F}$;
- (ii) If $A \in \mathcal{F}$, then $B \in \mathcal{F}$ for every $B \supset A$;
- (iii) If $A, B \in \mathcal{F}$, then $A \cap B \in \mathcal{F}$.

Clearly $\{X\}$ is a filter in X , so there exists a filter for any X . In addition, $\{X\}$ is the smallest filter we can ever make. Then, is there exists a maximal filter in X with respect to the set theoretical inclusion? Below proposition shows that indeed such filter exists, so this proposition justifies that Definition A.1.3 is well-defined.

Proposition A.1.2

Let X be a nonempty set and $\mathcal{F}$ be a filter in X . Then there exists a maximal filter $\mathcal{U} \supset \mathcal{F}$ in X meaning that there is no filter $\mathcal{G}$ in X which contains $\mathcal{U}$.

proof. Fix a filter $\mathcal{F}$ in X and define an ordered set

$$\mathcal{C} := \{\mathcal{B} \supset \mathcal{F} : \mathcal{B} \text{ is a filter in } X\}$$

with an inclusion order. Then by the Hausdorff maximal theorem, there exists a maximal totally ordered subset $\mathcal{D}$ of $\mathcal{C}$. Claim that $\bigcup \mathcal{D}$ is a filter and $\bigcup \mathcal{D} \supset \mathcal{F}$. Notice that the second one is trivial, so we are going to focus the first assertion.

Clearly $\emptyset \notin \bigcup \mathcal{D}$ and if $E \in \bigcup \mathcal{D}$ and $F \supset E$, then $F \in \bigcup \mathcal{D}$. If $E_1, E_2 \in \bigcup \mathcal{D}$, we can take two filters $\mathcal{F}_1, \mathcal{F}_2 \in \mathcal{D}$ such that $E_1 \in \mathcal{F}_1$ and $E_2 \in \mathcal{F}_2$, respectively. Since $\mathcal{D}$ is totally ordered, it suffices to assume that $\mathcal{F}_1 \subset \mathcal{F}_2$. In this case, we have $E_1 \cap E_2 \in \mathcal{F}_2 \subset \bigcup \mathcal{D}$, proving that $\bigcup \mathcal{D}$ is a filter.

Notice that $\bigcup \mathcal{D} \in \mathcal{C}$ so $\bigcup \mathcal{D}$ is the desired one, otherwise we can take a filter $\mathcal{G} \supset \bigcup \mathcal{D}$ but this makes a contradiction of the maximality of $\mathcal{D}$. $\square$

Definition A.1.3 (Ultrafilters)

Let X be a set. An *ultrafilter* $\mathcal{U}$ in X is a maximal filter.

Remark A.1.4

Proposition A.1.2 is so-called as an *ultrafilter lemma*. The ultrafilter lemma is weaker than the axiom of choice. This means that one cannot prove the axiom of choice by using the ultrafilter lemma.

As we know that such definition is quite difficult to draw some image what ultrafilters look like. There is an equivalent statement that describes ultrafilters.

Proposition A.1.5

Let X be a nonempty set and $\mathcal{U}$ be a filter in X . Then $\mathcal{U}$ is an ultrafilter if and only if for any $A \subset X$, either A or $X \setminus A$ belongs to $\mathcal{U}$.

proof. To prove the necessity, assume that there exists a subset $A \subset X$ such that both A and $X \setminus A$ are not in $\mathcal{U}$. Then the collection

$$\mathcal{U}' := \mathcal{U} \cup \{A \cap E : E \in \mathcal{U}\}$$

is a filter which contains $\mathcal{U}$ by Exercise A.1.6. This makes a contradiction because $\mathcal{U}$ is an ultrafilter.

Conversely, suppose that $\mathcal{U}$ is a filter which satisfies the property in above. If $\mathcal{U}$ is not maximal, then there exists an ultrafilter $\mathcal{U}' \supset \mathcal{U}$ by Proposition A.1.2. We are going to finish the proof by showing that $\mathcal{U} = \mathcal{U}'$.

Fix $A \in \mathcal{U}'$. By the property of $\mathcal{U}$, either A or $X \setminus A$ is contained in $\mathcal{U}$. In here, if $X \setminus A \in \mathcal{U}$ then since $\mathcal{U} \subset \mathcal{U}'$, we have $X \setminus A \in \mathcal{U}'$ hence

$$\emptyset = A \cap (X \setminus A) \in \mathcal{U}'$$

because $\mathcal{U}'$ is a filter, but this is nonsense. This derives to have $A \in \mathcal{U}$, therefore $\mathcal{U}' \subset \mathcal{U}$. This finishes the proof. $\square$

Exercise A.1.6*

Prove that the collection $\mathcal{U}'$ defined in Proposition A.1.5 is a filter.

Remark A.1.7

If $\mathcal{U}$ is an ultrafilter in X for some nonempty set X , clearly both A and $X \setminus A$ do not lie in $\mathcal{U}$ for any $A \subset X$, otherwise by the definition of the filter,

$$\emptyset = A \cap (X \setminus A) \in \mathcal{U},$$

which is a contradiction.

We have learned what is a filter and what is an ultrafilter. To deal with filters, we need to know how we can make some appropriate filter. If X be a nonempty set and $\mathcal{C}$ a some collection of subsets of X , then we can generate a filter $\mathcal{F}$ in X which contains $\mathcal{C}$.

Definition A.1.8 (Finite intersection property)

Let X be a set. A collection $\mathcal{C}$ of subsets of X has a *finite intersection property* if for any finite set $\mathcal{F} \subset \mathcal{C}$, the intersection $\bigcap \mathcal{F}$ is not empty. We also use the abbreviation FIP for the term “finite intersection property”.

Proposition A.1.9

Let X be a nonempty set and $\mathcal{C}$ a collection of subsets of X that has FIP. Then there exists a filter $\mathcal{F} \supset \mathcal{C}$ in X .

proof. We can find one of such filter explicitly:

$$\mathcal{F} = \{E \subset X : E \supset \bigcap \mathcal{D}, \mathcal{D} \text{ a finite subset of } \mathcal{C}\}. \quad (\text{A.1})$$

Since $\mathcal{C}$ has FIP, the emptyset does not contained in $\mathcal{F}$. It is clear that both $E \subset \mathcal{F}$ and $F \supset E$ imply $F \in \mathcal{F}$ by the definition of $\mathcal{F}$.

Fix $E_1, E_2 \in \mathcal{F}$ and take finite subsets $\mathcal{D}_1, \mathcal{D}_2 \subset \mathcal{C}$ such that $\bigcap \mathcal{D}_1 \subset E_1$ and $\bigcap \mathcal{D}_2 \subset E_2$, respectively. By the basic set theory knowledge, we obtain $E_1 \cap E_2 \supset \bigcap (\mathcal{D}_1 \cup \mathcal{D}_2)$ and notice that $\mathcal{D}_1 \cup \mathcal{D}_2$ is a finite subset of $\mathcal{C}$, implying that $E_1 \cap E_2 \in \mathcal{F}$. Therefore, $\mathcal{F}$ is a filter. It is clear by definition that $\mathcal{F} \supset \mathcal{C}$. $\square$

Sometimes, we will deal with not only filters but their subsets which acts like a filter. Thanks to Proposition A.1.9 we can consider the following objects.

Definition A.1.10 (Filter base)

Let X be a set. A *filter base* $\mathcal{B}$ is a nonempty collection of subsets of X such that

- (a) If $A, B \in \mathcal{B}$, then there exists $N \in \mathcal{B}$ such that $N \subset A \cap B$.
- (b) $\emptyset \notin \mathcal{B}$.

If $\mathcal{B}$ is a filter base, then the filter $\mathcal{F}$ generated by eq. (A.1) substituted of $\mathcal{B}$ for $\mathcal{C}$ is called *generated filter* from $\mathcal{B}$.

Remark A.1.11

Clearly every filter base satisfies FIP, so the generated filter is well-defined by Proposition A.1.9. Every filter is clearly a filter base and the generated filter is itself. By the property (a) in Definition A.1.10, eq. (A.1) and the below filter are identical.

$$\mathcal{F} = \{E \subset X : E \supset F, F \in \mathcal{B}\}.$$

After this definition, we will see why sequences can be replaced into filters in the general topology theories. This definition explains the way how filters are transformed by functions.

Definition A.1.12 (Pushforward and pullback of filters)

Let X and Y be nonempty sets and $f : X \rightarrow Y$ a function. If $\mathcal{F}$ is a filter in X , the filter

$$f_*\mathcal{F} := \{E \subset Y : E \supset f[F] : F \in \mathcal{F}\}$$

is called a *pushforward of a filter* $\mathcal{F}$. If $\mathcal{B}$ is a filter base in X , then the filter base

$$f_*\mathcal{B} := \{f[F] : F \in \mathcal{B}\}$$

is called a *pushforward of a filter base* $\mathcal{B}$.

If f is surjective and $\mathcal{F}$ is a filter in Y , the filter

$$f^*\mathcal{F} := \{E \subset X : \{f \in F\} : F \in \mathcal{F}\}$$

is called a *pullback of a filter* $\mathcal{F}$. If $\mathcal{B}$ is a filter base and the surjectivity of f may drop, then the filter base

$$f^*\mathcal{B} := \{\{f \in F\} : F \in \mathcal{B}\}$$

is called a *pullback of a filter base* $\mathcal{B}$.

The reader should check that Definition A.1.12 makes sense.

Exercise A.1.13

Prove that $f_*\mathcal{F}$ and $f^*\mathcal{F}$ defined in Definition A.1.12 are really a filter, respectively, show that $f_*\mathcal{B}$ and $f^*\mathcal{B}$ are really a filter base, respectively.

Remark A.1.14

If the surjectivity is dropped, then $f^*\mathcal{F}$ might not be a filter if $\mathcal{F}$ is a filter in Y .

The pushforward of a filter has a remarkable property: every pushforward of ultrafilters is again an ultrafilter. Notice that such property does not hold for pullbacks in general.

Proposition A.1.15

Let X, Y be nonempty sets, $f : X \rightarrow Y$ a function, and $\mathcal{U}$ an ultrafilter in X . Then $f_*\mathcal{U}$ is an ultrafilter in Y .

proof. Since $f_*\mathcal{U}$ is a filter, it suffices that it satisfies the sufficiency condition of Proposition A.1.5. Let $A \subset Y$ be fixed. Since $\mathcal{U}$ is an ultrafilter, by Proposition A.1.5, either $\{f \in A\} \in \mathcal{U}$ or $X \setminus \{f \in A\} \in \mathcal{U}$ holds. As $X \setminus \{f \in A\} = \{f \in Y \setminus A\}$, we can assume that $\{f \in A\} \in \mathcal{U}$. In this case, since $A \supset f[\{f \in A\}]$ and $f[\{f \in A\}] \in f_*\mathcal{U}$, we therefore have $A \in f_*\mathcal{U}$. The proof of this theorem is now completed by applying Proposition A.1.5. $\square$

Now let X be a topological space. Then we can define a convergence of filters. Here is the definition.

Definition A.1.16 (Convergence of filters)

Let X be a topological space. A filter $\mathcal{F}$ in X *converges* to $x \in X$ if every open neighborhood of x belongs to $\mathcal{F}$. In this case, we write $\mathcal{F} \rightarrow x$. A filter base $\mathcal{B}$ in X *converges* to $x \in X$ if the generated filter converges to x . In this case, we write $\mathcal{B} \rightarrow x$.

Below lemma tells us the equivalent statement of the convergence of a filter base. It will be useful to test the convergence.

Lemma A.1.17

Let X be a topological space and $\mathcal{B}$ is a filter base in X . Then $\mathcal{B} \rightarrow x \in X$ if and only if for every open neighborhood U of x , there exists $V \in \mathcal{B}$ such that $V \subset U$.

proof. If $\mathcal{B} \rightarrow x$, then by the definition the generated filter $\mathcal{F}$ converges to x , and this means $\mathcal{N}_x$ is contained in $\mathcal{F}$. Then each $U \in \mathcal{N}_x$ associates an element $V \in \mathcal{B}$ such that $V \subset U$ by the definition of the generated filter.

Conversely, if $\mathcal{B} \nrightarrow x$ and $\mathcal{F}$ a generated filter of $\mathcal{B}$, there is an open neighborhood U of x that does not belong to $\mathcal{F}$. Again, by the definition of the generated filter, this implies that $V \not\subset U$ for every $V \in \mathcal{B}$. $\square$

Remark A.1.18

One can ask about that if $x \in X$ is fixed then does there exist a filter converges to x . Recall that $\mathcal{N}_x$ is the collection of all open neighborhoods of x . Then one can easily checked that $\mathcal{N}_x$ is a filter base and $\mathcal{N}_x \rightarrow x$.

We are ready to see that why filters can be used instead of sequences in arbitrary topological spaces. The theorem uses a filter bases instead of filters. But recall that every filter is a filter base.

Theorem A.1.19

Let X, Y be topological spaces. Then

- (a) For any nonempty set $A \subset X$, $x \in \overline{A}$ if and only if there is a filter base $\mathcal{B}$ such that $A \in \mathcal{B}$ and $\mathcal{B} \rightarrow x$.
- (b) Let $f : X \rightarrow Y$ be a function. Then f is continuous at $x \in X$ if and only if for every filter base $\mathcal{B}$ in X with $\mathcal{B} \rightarrow x$, $f_*\mathcal{B} \rightarrow f(x)$ holds.

proof. (a) If $x \in \overline{A}$, then $N \cap A \neq \emptyset$ for every open neighborhood N of x . Now consider

$$\mathcal{C} := \{A\} \cup \mathcal{N}_x.$$

Then notice that $\mathcal{C}$ satisfies FIP because every finite intersection of open sets is also open. Thus Proposition A.1.9 gives a filter $\mathcal{F}$ that contains $\mathcal{C}$. Observe that by the definition of $\mathcal{C}$, $A \in \mathcal{F}$ and $\mathcal{F} \rightarrow x$. Finally, recall that $\mathcal{F}$ is clearly a filter base.

Conversely, suppose that there is a point $x \in X$ and a filter base $\mathcal{B}$ in X such that $A \in \mathcal{B}$ and $\mathcal{B} \rightarrow x$. In order to show $x \in \overline{A}$, it suffices to show that $N \cap A$ is not empty for every open neighborhood N of x . For fixed $N \in \mathcal{N}_x$, Lemma A.1.17 gives a set $V \in \mathcal{B}$ such that $V \subset N$. Since $\mathcal{B}$ is a filter base and $A \in \mathcal{B}$, we have

$$\emptyset \neq A \cap V \subset A \cap N.$$

This proves $x \in \overline{A}$ as N can be any open neighborhood of x .

- (b) Let f be continuous at $x \in X$ and $\mathcal{B}$ a filter base in X that converges to x . Fix any open neighborhood N of $f(x)$. Being continuous, there exists an open set U containing x such that $f[U] \subset N$. Notice that there exists $V \in \mathcal{B}$ such that $V \subset U$ by Lemma A.1.17. Then we have

$$N \supset f[U] \supset f[V].$$

By the definition of the pushforward of a filter base, $f[V] \in f_*\mathcal{B}$. Hence Lemma A.1.17 shows that $f_*\mathcal{B} \rightarrow f(x)$.

Conversely, suppose that f is not continuous at x . Then there exists an open set $W \ni f(x)$ such that no open neighborhood U of x satisfies $f[U] \subset W$. Claim that $f_*\mathcal{N}_x \not\rightarrow f(x)$. Since clearly $\mathcal{N}_x \rightarrow x$, it suffices to prove the claim to finish the proof.

By Lemma A.1.17, in order that $f_*\mathcal{N}_x \rightarrow f(x)$, there should exist an element $E \in f_*\mathcal{N}_x$ such that $E \subset W$. Notice that every element of $f_*\mathcal{N}_x$ is the type

$$f[U], \quad \forall U \in \mathcal{N}_x.$$

However we just take an open set W such that no open neighborhood U of x satisfies $f[U] \subset W$. This proves the claim and therefore the theorem. $\square$

If X is a metric space and $A \subset X$, then we know that $x \in \overline{A}$ if and only if there exists a sequence $(x_n)_n$ in A that converges to x as discussed in the introduction of this section. In Theorem A.1.19, $A \in \mathcal{F}$ corresponds to the sentence $(x_n)_n \subset A$, and clearly $\mathcal{F} \rightarrow x$ means $x_n \rightarrow x$ as $n \rightarrow \infty$. In addition, statement Theorem A.1.19 (ii) is quite clear to explain whether a function is continuous at

x by considering the assertion related to a sequence.

Furthermore, note that on a Hausdorff space, every sequence converges at most one point. This is also true for filters, and we can say more.

Theorem A.1.20

Let X be a topological space. Then X is Hausdorff if and only if every filter base converges at most one point. That is, if a filter base $\mathcal{B}$ in X converges in both $x \in X$ and $y \in X$, then $x = y$.

proof. Let X be Hausdorff and assume that there exists a filter base $\mathcal{B}$ converges to different points x and y . Take two disjoint open sets U and V such that $x \in U$ and $y \in V$. Since $\mathcal{B} \rightarrow x$, by Lemma A.1.17, there exists $W_1 \in \mathcal{B}$ such that $W_1 \subset U$. Similarly, as $\mathcal{B} \rightarrow y$, we can take $W_2 \in \mathcal{B}$ satisfies $W_2 \subset V$. Since $\mathcal{B}$ is a filter base, $W_1 \cap W_2$ cannot be empty. However, we find

$$W_1 \cap W_2 \subset U \cap V = \emptyset,$$

which is a contradiction. Hence, there is no such filter base if X is Hausdorff.

Conversely, if X is not Hausdorff, then we can take two distinct points x and y such that there is no pair (U, V) of open sets such that $x \in U$, $y \in V$, and $U \cap V = \emptyset$. Then define

$$\mathcal{B} := \{F \cap G : F \in \mathcal{N}_x, G \in \mathcal{N}_y\}$$

By the choice of x and y , $\mathcal{B}$ does not have an emptyset as its element, so it is easy to show that $\mathcal{B}$ is a filter base. Notice by the definition of $\mathcal{B}$ that $\mathcal{B} \rightarrow x$ and $\mathcal{B} \rightarrow y$. This proves the theorem. $\square$

A.2 Compact sets

We already see the definition of compact sets and their properties by using nets. In this section, we study the same theorems but in filters.

Theorem A.2.1

Let X be a topological space. Then X is compact if and only if every ultrafilter in X converges.

proof. Suppose that there is an ultrafilter $\mathcal{U}$ which converges nowhere in X . For each $x \in X$, take an open set U_x which is not contained in $\mathcal{U}$. Note that clearly $\{U_x\}_{x \in X}$ is an open cover of X . If X were compact, there are finitely many points $x_1, \dots, x_n$ such that

$$U_{x_1} \cup U_{x_2} \cup \dots \cup U_{x_n} \supset X. \quad (\text{A.2})$$

However as $\mathcal{U}$ is an ultrafilter, $X \setminus U_x \in \mathcal{U}$ holds for each $x \in X$ by Proposition A.1.5. This and eq. (A.2) gives

$$\emptyset = (X \setminus U_{x_1}) \cap (X \setminus U_{x_2}) \cap \dots \cap (X \setminus U_{x_n}) \in \mathcal{U},$$

which is a contradiction. Such contradiction occurs as we assumed that X is compact. Hence X cannot be compact.

Conversely, suppose that X is not compact. Then there is an open cover $\mathcal{U}$ which does not have a finite subset of $\mathcal{U}$ that covers X . Put

$$\mathcal{V} := \{X \setminus E : E \in \mathcal{U}\}.$$

Observe that $\mathcal{V}$ satisfies FIP. Take an ultrafilter $\mathcal{U}$ which is a superset of $\mathcal{V}$ by Proposition A.1.2 and Proposition A.1.9. Since $\mathcal{V} \subset \mathcal{U}$ and $\mathcal{U}$ is an open cover of X , there is no $x \in X$ such that

$\mathcal{U} \rightarrow x$. Indeed, if $x \in X$ then there is $U \in \mathcal{U}$ that contains x . Since $X \setminus U \in \mathcal{U}$, U does not belong to $\mathcal{U}$ and this implies that $\mathcal{U}$ does not converges to x . $\square$

Corollary A.2.2

Let X be a topological space. Then nonempty subset $K \subset X$ is compact if and only if every ultrafilter $\mathcal{U}$ in X containing K converges to some point in K . Notice that the convergent point can differ for each ultrafilter.

proof. Suppose that $K \subset X$ is compact, and fix an ultrafilter $\mathcal{U}$ that contains K . Then the set

$$\mathcal{U}' := \{F \cap K : F \in \mathcal{U}\} \quad (\text{A.3})$$

is an ultrafilter in K by Exercise A.2.3. Being compact, $\mathcal{U}' \rightarrow x \in K$ by ???. Now fix an open set N which contains x . Since $N \cap K$ is open in K and $x \in N \cap K$, we have $N \cap K \in \mathcal{U}'$ because $\mathcal{U}'$ converges to x in K .

In order to prove that $N \in \mathcal{U}$, assume that N is not contained in $\mathcal{U}$. Then by Proposition A.1.5, $X \setminus N$ is contained in $\mathcal{U}$, thus $(X \setminus N) \cap K \in \mathcal{U}'$. Since $N \cap K$ is also contained in $\mathcal{U}'$, and $\mathcal{U}'$ is a filter (actually an ultrafilter), we must obtain

$$\emptyset = (N \cap K) \cap [(X \setminus N) \cap K] \in \mathcal{U}',$$

which is a contradiction. Therefore, $N \in \mathcal{U}$, and thus we prove that $\mathcal{U} \rightarrow x$ since N is arbitrary.

To prove the sufficiency, claim that if $\mathcal{U}'$ is any ultrafilter in K , then there exists an ultrafilter $\mathcal{U}$ in X such that both $\mathcal{U}$ and $\mathcal{U}'$ satisfies eq. (A.3). Since $\mathcal{U}'$ satisfies FIP, by Proposition A.1.9 and Proposition A.1.2, there exists an ultrafilter $\mathcal{U}$ which contains $\mathcal{U}'$. This proves that

$$\mathcal{U}' \subset \{F \cap K : F \in \mathcal{U}\} =: \tilde{\mathcal{U}}.$$

On the other hand, as $\mathcal{U}$ is an ultrafilter which contains K because we take $\mathcal{U}$ satisfies $\mathcal{U} \supset \mathcal{U}'$, by Exercise A.2.3, $\tilde{\mathcal{U}}$ is an ultrafilter in K . Since $\mathcal{U}'$ is also an ultrafilter in K which is contained in another filter $\tilde{\mathcal{U}}$, they must be identical by the definition of an ultrafilter. Hence, we proved that $\mathcal{U}$ and $\mathcal{U}'$ satisfies eq. (A.3).

From this fact, we are ready to prove that K is compact provided that every ultrafilter in X containing K converges to some point in K . Fix any ultrafilter $\mathcal{U}'$ in K and take an ultrafilter $\mathcal{U}$ in X satisfies eq. (A.3). By our assumption, $\mathcal{U}$ converges to some $x \in K$. By eq. (A.3) and the definition of the subspace topology, it is easy to verify that $\mathcal{U}' \rightarrow x$. Therefore, K is compact because of ??. $\square$

Exercise A.2.3*

Prove that eq. (A.3) is really an ultrafilter in K .

Below propositions can be proved without using any filter theory. In here, however, we will see how ? (and ?) can be used to prove several compactness properties. One may solve ?.

Proposition A.2.4

Let X be a topological space and $K \subset X$ be a nonempty set.

- (a) If X is Hausdorff and K is compact, then K is closed.
- (b) If X is compact and K is closed, then K is compact.

- (c) Let Y be another topological space and $f : X \rightarrow Y$ a continuous map, then $f[K]$ is compact in Y whenever K is compact in X .
- (d) Let X be Hausdorff and $\mathcal{K}$ be a nonempty collection of compact sets which satisfies FIP. Then $\bigcap \mathcal{K}$ is not empty.

proof. (a) Fix $x \in \overline{K}$ and take a filter $\mathcal{F}$ in X such that $K \in \mathcal{F}$ and $\mathcal{F} \rightarrow x$ by Theorem A.1.19. Then take an ultrafilter $\mathcal{U} \supset \mathcal{F}$ by Proposition A.1.2. Since $\mathcal{F} \rightarrow x$ and $\mathcal{U}$ contains $\mathcal{F}$, we have $\mathcal{U} \rightarrow x$. On the other hand, since K is compact, and $K \in \mathcal{F} \subset \mathcal{U}$, by ??, there exists $y \in K$ such that $\mathcal{U} \rightarrow y$. Since X is Hausdorff, x and y must be equal because of Theorem A.1.20, proving that $x = y \in K$. Since x is arbitrary element of $\overline{K}$, we proved that K is closed.

- (b) Fix an ultrafilter $\mathcal{U}$ in X which contains K . Being compact, $\mathcal{U}$ converges to some point $x \in X$. By Theorem A.1.19, $x \in \overline{K} = K$. Hence, K is compact because of ?? as $\mathcal{U}$ is arbitrary.
- (c) Fix an ultrafilter $\mathcal{U}$ in $f[K]$. Then take an ultrafilter $\mathcal{U}'$ in K such that $f_*\mathcal{U} \subset \mathcal{U}'$. Being compact, we can take a point $x \in K$ such that $\mathcal{U}' \rightarrow x$ by ??. Since f is continuous, $f_*\mathcal{U}' \rightarrow f(x)$ because of Theorem A.1.19. Observe that $\mathcal{U} \subset f_*\mathcal{U}'$, hence both should be same since $\mathcal{U}$ is an ultrafilter. Therefore, $\mathcal{U}$ converges to $f(x)$, and we proved that $f[K]$ is compact by ?? as $\mathcal{U}$ is arbitrary.
- (d) Since $\mathcal{K}$ satisfies FIP, Proposition A.1.9 and Proposition A.1.2 gives an ultrafilter $\mathcal{U}$ in X such that $\mathcal{U} \supset \mathcal{K}$. Notice that $\mathcal{K}$ is a collection of compact sets. Thus by ?? each compact set $K \in \mathcal{K}$ associates a point $x_K \in K$ such that $\mathcal{U} \rightarrow x_K$. As X is Hausdorff, the ultrafilter $\mathcal{U}$ converges at most one point by Theorem A.1.20. Therefore, points x_K are actually identical, say x . Since $x = x_K \in K$ for every $K \in \mathcal{K}$, we find a point x which belongs to $\bigcap \mathcal{K}$.

This proves the proposition. $\square$

Exercise A.2.5

Prove Proposition A.2.4 *without using any filter theories*.

A.3 Another proof of Tychonoff's Theorem

We already studied the Tychonoff's theorem using a filter. In here, we are going to prove the same theorem but in net.

Theorem A.3.1 (Tychonoff's Theorem)

Let $\{X_\alpha\}_{\alpha \in A}$ be a collection of compact topological spaces where A is an index set. Then the space $X = \prod_{\alpha \in A} X_\alpha$ equipped with the product topology is compact.

proof. Let us define a projection $\pi_\alpha : X \rightarrow X_\alpha$ for each $\alpha \in A$, and fix a universal net $(x_\lambda)_{\lambda \in \Lambda}$ in X . By ??, $(\pi_\alpha)_*(x_\lambda)$ is a universal net in X_α , so we can take a point $z_\alpha \in X_\alpha$ such that $(\pi_\alpha)_*(x_\lambda) \rightarrow z_\alpha$. Such points exists because X_α is compact and recall Theorem 1.6.2. Finally, take the point $x \in X$ such that $\pi_\alpha(x) = z_\alpha$ for every $\alpha \in A$. Claim that $x_\lambda \rightarrow x$.

To prove this, fix an open neighborhood N of x in X . Since

$$\{\pi_\alpha^{-1}[U] : \alpha \in A, U \text{ open in } X_\alpha\}$$

forms a subbasis of X , there exists $\alpha_1, \dots, \alpha_n \in A$, and $U_{\alpha_1}, \dots, U_{\alpha_n}$ are open sets in X_{α_j} , $j = 1, 2, \dots, n$, respectively, such that

$$x \in \pi_{\alpha_1}^{-1}[U_{\alpha_1}] \cap \pi_{\alpha_2}^{-1}[U_{\alpha_2}] \cap \dots \cap \pi_{\alpha_n}^{-1}[U_{\alpha_n}] \subset N.$$

Notice that

$$z_{\alpha_j} \in U_{\alpha_j}$$

for $j = 1, 2, \dots, n$. This gives that U_{α_j} eventually contains $(\pi_{\alpha_j})_*(x_\lambda)$, hence $\pi_{\alpha_j}^{-1}[U_{\alpha_j}]$ eventually contains $(x_\lambda)_{\lambda \in \Lambda}$.

Thus,

$$\begin{aligned} (x_\lambda)_{\lambda \in \Lambda} \text{ is eventually in } \pi_{\alpha_1}^{-1}[U_{\alpha_1}] \cap \pi_{\alpha_2}^{-1}[U_{\alpha_2}] \cap \dots \cap \pi_{\alpha_n}^{-1}[U_{\alpha_n}] \\ \Rightarrow N \text{ eventually contains } (x_\lambda)_{\lambda \in \Lambda}. \end{aligned}$$

Since N is arbitrary open neighborhood of x , we have proved that $x_\lambda \rightarrow x$. Therefore, X is compact because of Theorem 1.6.2. $\square$

A.4 Relation between nets and filters

We now learned both filters and nets. As it is mentioned in earlier, both concepts are made to generalize sequences. Then one might question about the correspondence between filters and nets. Indeed, one can canonically construct a filter from a net, and vice versa, preserving their convergence properties seamlessly.

Definition A.4.1 (Eventuality filters and canonical nets)

Let X be a nonempty set.

- (i) If $(x_\alpha)_{\alpha \in A}$ is a net in X , the collection of all sets $E \subset X$ such that $(x_\alpha)_{\alpha \in A}$ is eventually in E forms a filter. We call this the *eventuality filter* of the net $(x_\alpha)_{\alpha \in A}$, and denote it by $\mathcal{F}(x_\alpha)$.
- (ii) If $\mathcal{F}$ is a filter in X , define a set $D_{\mathcal{F}} = \{(x, F) : F \in \mathcal{F}, x \in F\}$. Equip $D_{\mathcal{F}}$ with a preorder $\leq$ defined by $(x, F) \leq (y, G)$ if $G \subset F$. Then $D_{\mathcal{F}}$ is a directed set. The net $N_{\mathcal{F}}$ defined by $N_{\mathcal{F}}(x, F) = x$ from $D_{\mathcal{F}}$ to X is called the *canonical net* of the filter $\mathcal{F}$.

Exercise A.4.2

Prove that $\mathcal{F}(x_\alpha)$ is indeed a filter in X , and that $D_{\mathcal{F}}$ is a directed set.

Theorem A.4.3

Let X be a topological space and $x \in X$.

- (a) A net $(x_\alpha)_{\alpha \in A}$ converges to x if and only if its eventuality filter $\mathcal{F}(x_\alpha)$ converges to x .
- (b) A filter $\mathcal{F}$ converges to x if and only if its canonical net $N_{\mathcal{F}}$ converges to x .
- (c) A point x is a cluster point of $(x_\alpha)_{\alpha \in A}$ if and only if $x \in \bigcap_{E \in \mathcal{F}(x_\alpha)} \overline{E}$.

proof. (a) By definition, a net $(x_\alpha)_{\alpha \in A}$ converges to x if and only if it is eventually in every open neighborhood U of x . This is exactly equivalent to saying that every open neighborhood U of x belongs to $\mathcal{F}(x_\alpha)$, which means $\mathcal{F}(x_\alpha) \rightarrow x$.

(b) Suppose $\mathcal{F} \rightarrow x$. For any open neighborhood U of x , $U \in \mathcal{F}$. Take any $y \in U$. Then $(y, U) \in D_{\mathcal{F}}$. For any $(z, G) \geq (y, U)$, we have $G \subset U$, hence $z \in G \subset U$. This implies the net $N_{\mathcal{F}}$ is eventually in U , so $N_{\mathcal{F}} \rightarrow x$.

Conversely, if $N_{\mathcal{F}} \rightarrow x$, then for any open neighborhood U of x , there exists $(y, F) \in D_{\mathcal{F}}$ such that for all $(z, G) \geq (y, F)$, $z \in U$. In particular, for any $z \in F$, we have $(z, F) \geq (y, F)$, meaning $z \in U$. Thus $F \subset U$. Since $F \in \mathcal{F}$ and $\mathcal{F}$ is a filter, $U \in \mathcal{F}$. Therefore, $\mathcal{F} \rightarrow x$.

(c) Suppose x is a cluster point of $(x_{\alpha})_{\alpha \in A}$. This means for any $U \in \mathcal{N}_x$ and any $\alpha \in A$, there is $\gamma \geq \alpha$ such that $x_{\gamma} \in U$. By the definition of $\mathcal{F}(x_{\alpha})$, every $E \in \mathcal{F}(x_{\alpha})$ contains a tail set $T_{\alpha} = \{x_{\gamma} : \gamma \geq \alpha\}$ for some α . Thus $U \cap E \supset U \cap T_{\alpha} \neq \emptyset$. Since U is arbitrary, $x \in \overline{E}$, hence $x \in \bigcap_{E \in \mathcal{F}(x_{\alpha})} \overline{E}$.

Conversely, if $x \in \bigcap_{E \in \mathcal{F}(x_{\alpha})} \overline{E}$, then for any $U \in \mathcal{N}_x$ and $\alpha \in A$, since the tail $T_{\alpha} \in \mathcal{F}(x_{\alpha})$, we have $x \in \overline{T_{\alpha}}$, which implies $U \cap T_{\alpha} \neq \emptyset$. This means there is $\gamma \geq \alpha$ with $x_{\gamma} \in U$. Thus x is a cluster point. $\square$

Not only the convergence, but the maximality concepts (universal nets and ultrafilters) also perfectly correspond to each other.

Theorem A.4.4

Let X be a set.

- (a) If $(x_{\alpha})_{\alpha \in A}$ is a universal net in X , then $\mathcal{F}(x_{\alpha})$ is an ultrafilter in X .
- (b) If $\mathcal{U}$ is an ultrafilter in X , then $N_{\mathcal{U}}$ is a universal net in X .

proof. (a) Suppose $(x_{\alpha})_{\alpha \in A}$ is a universal net. To show $\mathcal{F}(x_{\alpha})$ is an ultrafilter, by Proposition A.1.5, it suffices to show that for any $E \subset X$, either E or $X \setminus E$ belongs to $\mathcal{F}(x_{\alpha})$. Since the net is universal, it is eventually in E or eventually in $X \setminus E$ (??). By definition, this means either $E \in \mathcal{F}(x_{\alpha})$ or $X \setminus E \in \mathcal{F}(x_{\alpha})$, proving the claim.

(b) Suppose $\mathcal{U}$ is an ultrafilter. To show $N_{\mathcal{U}}$ is a universal net, we use ??. Let $E \subset X$. Since $\mathcal{U}$ is an ultrafilter, either $E \in \mathcal{U}$ or $X \setminus E \in \mathcal{U}$. Assume $E \in \mathcal{U}$. Fix any $y \in E$. For any $(z, G) \geq (y, E)$, we have $G \subset E$, so $z \in G \subset E$. This shows that $N_{\mathcal{U}}$ is eventually in E . By the same argument, if $X \setminus E \in \mathcal{U}$, $N_{\mathcal{U}}$ is eventually in $X \setminus E$. Thus $N_{\mathcal{U}}$ is a universal net. $\square$

Proofs of several lemmas

B.1 Proof of Lemma 2.4.5

If $f(a) = 0$, since f is nonnegative and decreasing, $f \equiv 0$ on $[a, b]$. Thus in this case, there is nothing to do, and it allows to assume that $f(a) > 0$.

Since f is Riemann integrable on $[a, b]$, for fixed $\epsilon > 0$ take a partition $P = \{a = x_0 < x_1 < \dots < x_n = b\}$ such that

$$\int_a^b f(x)dx < \frac{\epsilon}{2} + L(f, P), \quad (\text{B.1})$$

where $L(f, P)$ is a lower Riemann sum of f with respect to P . Since f is decreasing, one can calculate $L(f, P)$ explicitly:

$$L(f, P) = \sum_{j=1}^n (x_j - x_{j-1}) \left(\inf_{x \in [x_{j-1}, x_j]} f(x) \right) = \sum_{j=1}^n (x_j - x_{j-1}) f(x_j).$$

Now we define a nonnegative continuous function g on $[a, b]$ as follows. Define

$$\delta := \frac{\epsilon}{f(a)} \wedge \min_{1 \leq j \leq n} (x_j - x_{j-1}) > 0,$$

define $g(x) = f(b)$ for every $x \in [x_{n-1}, b]$, and construct g on $[x_{j-1}, x_j]$ where $j = 1, \dots, n-1$ by

(i) $g(x) = f(x_j)$ where $x \in [x_{j-1}, x_j - \delta]$;

(ii) g is defined by a straight line between points $(x_j - \delta, f(x_j))$ and $(x_j, f(x_{j+1}))$.

Since f is decreasing, for $j = 1, \dots, n-1$, values of g on $[x_{j-1}, x_j]$ is at most $f(x_j)$, which is clearly less or equal than f . In addition, by its construction, g is obviously nonnegative and continuous. Thus we have $0 \leq g \leq f$, and g is continuous.

It remains to calculate the integral of g on $[a, b]$, which is

$$\begin{aligned} \int_a^b g(x)dx &= \int_{x_{n-1}}^b g(x)dx + \sum_{j=1}^{n-1} \int_{x_{j-1}}^{x_j} g(x)dx \\ &= \int_{x_{n-1}}^b f(b)dx + \sum_{j=1}^{n-1} \left(\int_{x_{j-1}}^{x_j - \delta} f(x_j)dx + \int_{x_j - \delta}^{x_j} \frac{f(x_{j+1}) - f(x_j)}{\delta} (x - x_j + \delta) + f(x_j)dx \right) \\ &= f(b)(b - x_{n-1}) + \sum_{j=1}^{n-1} ((x_j - x_{j-1} - \delta)f(x_j) + \frac{\delta}{2}(f(x_j) + f(x_{j+1}))). \end{aligned}$$

Hence,

$$L(f, P) - \int_a^b g(x)dx = \frac{\delta}{2} \sum_{j=1}^{n-1} (f(x_j) - f(x_{j+1})) = \frac{\delta}{2} (f(x_1) - f(b)) \leq \frac{\delta}{2} f(a) \leq \frac{\epsilon}{2}.$$

Combining this and eq. (B.1), we obtain the desired conclusion. The proof is now finished. $\square$

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